CLICK HERE CLICK HERE

APUL

NEET FLIX

AIM FREE EDUCATION TO ALL

NEET CHANNEL

JOIN OUR TELEGRAM CHANNELS HERE

TEST PAPERS <<< CLICK

TOPPER'S NOTES <<< CLICK

BEST FROM KOTA LECTURES <<< CLICK

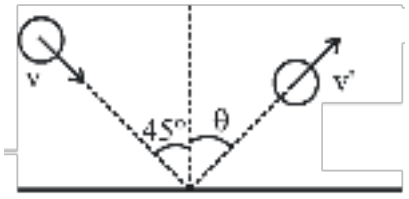
QUIZES <<< CLICK

REVISION FLASH CARDS &

MANY MORE <<< CLICK

PHYSICS

1) A ball collides to a surface as shown in figure. If coefficient of restitution $e = \frac{1}{\sqrt{3}}$, then θ will be:-

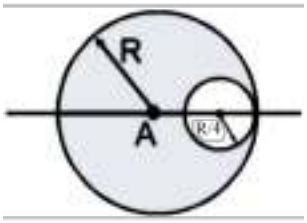


(1) 45°
(2) 60°
(3) 30°
(4) 0°

2) A monkey of mass 20 kg rides on a 40 kg trolley moving at a constant speed of 8 m/s along a horizontal track. Frictional force can be neglected between trolley and track. If the monkey jumps vertically, with respect to the moving trolley, to catch the overhanging branch of tree, the speed of the trolley after the monkey has jumped off is :-

(1) 16m/s
(2) 12m/s
(3) 8m/s
(4) 6m/s

3) The center of mass of the shaded portion of the disc is : (The mass is uniformly distributed in the shaded portion):



(1) $\frac{R}{20}$ to the left of A
(2) $\frac{R}{12}$ to the left of A
(3) $\frac{R}{20}$ to the right of A
(4) $\frac{R}{12}$ to the right of A

4) Two identical balls moving with a speed of 6 m/s in opposite direction, collides. If for collision,

AIM : FREE EDUCATION TO ALL ♥ APUL

$e = \frac{1}{3}$; the speed of each ball after collision will be :-

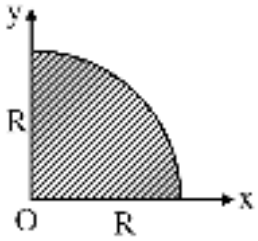
(1) 2 m/s

(2) 4 m/s

(3) 6 m/s

(4) 1 m/s

5) A quater sector of a uniform disc of radius R is shown in figure. The co-ordinates of centre of mass



are :

(1) $\left(\frac{2R}{3\pi}, \frac{2R}{3\pi}\right)$

(2) $\left(\frac{2R}{3\pi}, \frac{4R}{3\pi}\right)$

(3) $\left(\frac{4R}{3\pi}, \frac{4R}{3\pi}\right)$

(4) $\left(\frac{4R}{3\pi}, \frac{2R}{3\pi}\right)$

6) A rod of length L is hinged at one end and it is rotated with a constant angular velocity in a horizontal plane. Let T_1 and T_2 be the tensions at the points L/4 and 3L/4 away from the hinged end. Then :

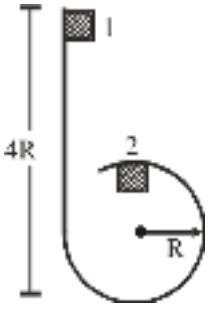
(1) $T_1 > T_2$

(2) $T_2 > T_1$

(3) $T_1 = T_2$

(4) The relation between T_1 and T_2 depends on whether the rod rotates clockwise or anticlockwise

7) A cube of mass M starts at rest from point 1 at a height 4R, where R is the radius of the circular track. The cube slides down the frictionless track and around the loop. The force which the track



exerts on the cube at point 2 is :

(1) 3 mg

(2) mg

AIM : FREE EDUCATION TO ALL APUL

(3) 2 mg

(4) cube will not reach the point 2.

8) **Assertion (A) :-** During a turn, the value of centripetal force should not be more than the limiting frictional force.
Reason (R) :- The centripetal force is provided by the frictional force between the tyres and the road.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

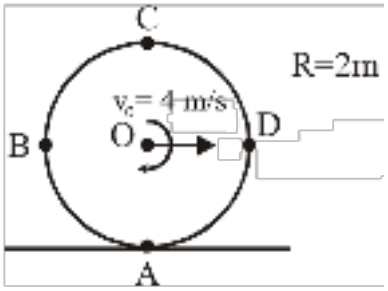
9) If $\theta = (4t^3 - 3t^2 + 2t + 1)$ radian then angular velocity and angular acceleration at t = 2 second are :

(1) 38 rad/s, 48 rad/s

(2) 42 rad/s^2, 32 rad/s^2

(3) 38 rad/s, 42 rad/s^2

(4) 42 rad/s^2, 32 rad/s

10) A disc rolls purely on ground as shown in figure: 

Which of following correct? (V_0 = velocity of point O)

(1) $v_A = 0$, $v_B = 4\sqrt{2}$ m/s

(2) $\omega = 2$ rad/s, $v_B = v_D$

(3) $v_C = 8$ m/s, $V_C = 2V_0$

(4) All of the above

11) A solid cylinder of mass 2 kg and radius 50 cm rolls up an inclined plane of angle of inclination 30°. The centre of mass of cylinder has speed of 4 m/s. The distance travelled by the cylinder on the incline surface will be : (Take g = 10 m/s^2)

(1) 2.2 m

(2) 1.6 m

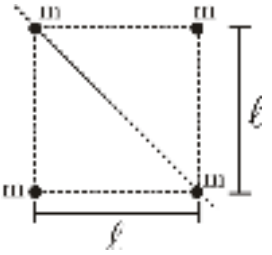
(3) 1.2 m

(4) 2.4 m

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL ❤ APUL

12) Find moment of inertia of system about diagonal of square :-



(1) $2m\square^2$

(2) $m\square^2$

(3) $\frac{m\ell^2}{2}$

(4) $\frac{m\ell^2}{\sqrt{2}}$

13) The radius of gyration of a rod of length L and mass M about an axis perpendicular to its length and passing through a point at a distance L/3 from one of its ends is :-

(1) $\frac{\sqrt{7}}{6}L$

(2) $\frac{L}{9}$

(3) $\frac{L}{3}$

(4) $\frac{\sqrt{5}}{2}L$

14) A boy and a man carry a uniform rod of length L, horizontally in such a way that the boy gets $\frac{1}{4}$th load. If the boy is at one end of the rod, the distance of the man from the other end is

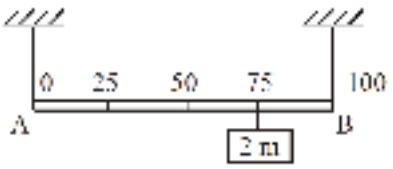
(1) $\frac{L}{3}$

(2) $\frac{L}{4}$

(3) $\frac{2L}{3}$

(4) $\frac{3L}{4}$

15)



Shown in the figure a rigid and uniform one meter long rod AB held in horizontal position by two strings tied to its ends and attached to the ceiling. The rod is of mass 'm' and has another weight of mass 2 m hung at a distance of 75 cm from A. The tension in the string at A is :

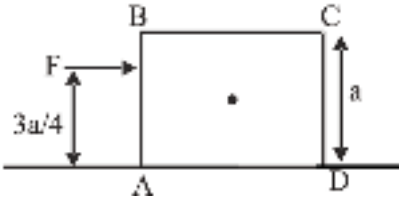
(1) 2 mg

(2) 0.5 mg

(3) 0.75 mg

(4) mg

16) A uniform cube of side 'a' and mass 'm' rests on a rough horizontal surface. A horizontal force F is applied normal to one of the faces at a point that is directly above the centre of the face, at a height $\frac{3a}{4}$ above the base. The minimum value of F for which the cube begins to topple about the edge is (Assume that the cube does not slide).



(1) $\frac{3}{2}$mg

(2) $\frac{2}{3}$mg

(3) $\frac{4}{5}$mg

(4) $\frac{1}{6}$mg

17) A rotating wheel has a speed of 1200 r.p.m. and then it is made to slow down at a constant rate at 2 rad/sec^2. The number of revolution it makes before coming to rest will be :-

(1) 272

(2) 628

(3) 143

(4) 314

18) A circular disc has a mass of 1 kg and radius 40 cm. It is rotating about an axis passing through its centre and prependicular to its plane with a speed of 10 rev/s. The work done in joules in stopping it would be :-

(1) 4

(2) 47.5

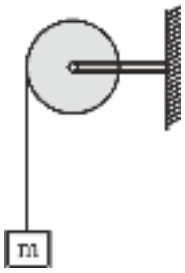
(3) 79

(4) 158

19) A disc of mass m and radius r is free to rotate about its centre as shown in the figure. A string is wrapped over its rim and a block of mass m is attached to the free end of the string. The system is

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL



released from rest. The speed of the block as it descends through a height h, is :-

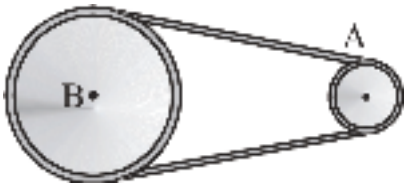
(1) $\sqrt{2gh}$

(2) $\sqrt{\frac{2}{3}gh}$

(3) $2\sqrt{\frac{gh}{3}}$

(4) $\frac{1}{2}\sqrt{3gh}$

20) Wheels A and B in Figure are connected by a belt that does not slip. The radius of B is 3.00 times the radius of A. What would be the ratio of the rotational inertias I_A/I_B if the two wheels had same angular momentum about their central axes :-



(1) $\frac{3}{4}$

(2) $\frac{2}{3}$

(3) $\frac{1}{3}$

(4) $\frac{1}{2}$

21) Match the column :

Column I		Column II	
A.	Gravitational potential at the centre of a solid sphere	**P.**	$\frac{dA}{dt}$ = Constant
B.	Escape velocity from earth surface	**Q.**	$-\frac{GMm}{2R}$
C.	Kepler's second law	**R.**	$-\frac{3}{2}\frac{GM}{R}$
D.	Total energy of satellite very close to earth surface	**S.**	11.2 km/s

(1) A-Q, B-S, C-P, D-R

(2) A-R, B-P, C-S, D-Q

(3) A-R, B-S, C-Q, D-P

(4) A-R, B-S, C-P, D-Q

22) By what amount, the energy of a satellite of mass m has to be increased to shift it from an orbit of radius r to orbit of radius $\frac{3}{2}r$? (mass of earth is M)

(1) $\frac{GMm}{r}$

(2) $\frac{GMm}{2r}$

(3) $\frac{GMm}{4r}$

(4) $\frac{GMm}{6r}$

23) The height from the surface of earth at which gravitational field intensity becomes 1% of its value on the surface of the earth is :-(R_e :- radius of earth)

(1) 100 R_e

(2) 10 R_e

(3) 99 R_e

(4) 9 R_e

24) A body weight 100N on surface of the earth. When it is taken to a depth $\frac{R}{4}$ from earth's surface, it would weight :-

(1) 100 N

(2) 25 N

(3) 50 N

(4) 75 N

25) If the mass of the earth is decreased by 1% and radius of earth is increased by 2% then find out percentage change in acceleration due to gravity on the surface of earth :-

(1) increases by 5%

(2) decreases by 5%

(3) increases by 4%

(4) decreases by 4%

26) The orbital angular momentum of a satellite revolving at a distance r from the center of earth is L. If the distance is increased from r to 64 r then the new angular momentum will be.

(1) 16 L

(2) 64 L

(3) $\frac{L}{4}$

(4) 8L

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL ❤ APUL

27) A particle is projected vertically upwards from earth surface with speed equal to one third of escape velocity then maximum height attained by it, is :- (R → Radius of earth)

(1) $\frac{R}{4}$

(2) $\frac{R}{8}$

(3) $\frac{R}{2}$

(4) 2R

28)

For a satellite revolving around earth the time period is T, then it's kinetic energy is proportional to :-

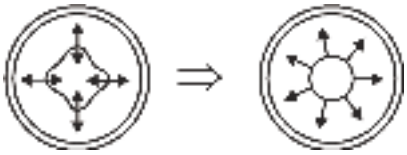
(1) T^{-2}

(2) $T^{-2/3}$

(3) T^{-1}

(4) T^{-3}

29) A soap film is formed on a circular frame. A loop of thread is lying on film. If the film inside the loop is broken, then tension in the thread will be :

(R = radius of loop, T = surface tension of liquid) 

(1) 2πRT

(2) πRT

(3) RT

(4) $\pi R^2 T$

30) An air bubble increases its area to four times when it comes up to the surface from bottom. The depth of lake is :- (P_0 = 1 atm = 10^5 Pa)

(1) 30 m

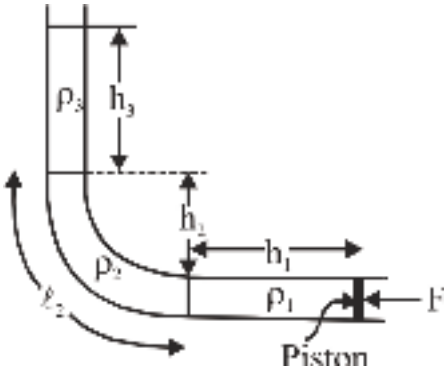
(2) 70 m

(3) 60 m

(4) None

31) A tube is bent into L shape and kept in a vertical plane. If these three liquids are kept in

AIM : FREE EDUCATION TO ALL APUL



equilibrium by the piston of area A, the value of $\frac{F}{A}$ is :

(1) $(\rho_1 h_1 + \rho_2 h_2 + \rho_3 h_3)g$

(2) $(\rho_1 h_1 + \rho_2 \square_2 + \rho_3 h_3)g$

(3) $(\rho_2 h_2 + \rho_3 h_3)g$

(4) $(\rho_2 \square_2 + \rho_3 h_3)g$

32) Water is being poured in a vessel at a constant rate α m^3/s. There is a small hole of area a at the bottom of the tank. The maximum level of the water in the vessel is proportional to :
(a) α (b) α^2 (c) a^{-1} (d) a^{-2}

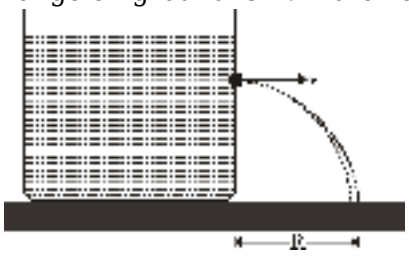
(1) c, b

(2) a, c

(3) b, d

(4) b, a

33) In the figure shown velocity of liquid which comes out is v, time of liquid to fall to ground is t and range on ground is R. If the vessel is taken to a high mountain, then match the following.



	Column I		**Column II**
(A)	v	(P)	will increase
(B)	t	(Q)	will decreases
(C)	R	(R)	will remain same

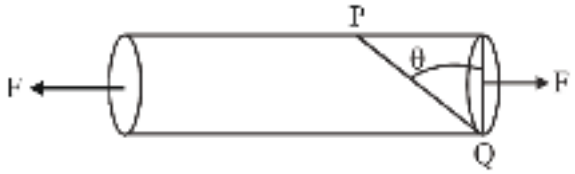
(1) A → P, B → Q, C → R

(2) A → Q, B → P, C → R

(3) A → R, B → P, C → Q

(4) A → Q, B → R, C → R

34) A force F is applied along a rod of transverse sectional area A. The normal stress to a section PQ



inclined θ to transverse section is:-

(1) $\frac{F\sin\theta}{A}$

(2) $\frac{F}{A}\cos\theta$

(3) $\frac{F}{2A}\sin 2\theta$

(4) $\frac{F}{A}\cos^2\theta$

35) The breaking stress of a material is 10^6 N/m^2. If the density of the material is 3×10^3 kg/m^3, what should be the length of the material so that it breaks by its own weight? (Take g = 10 m/s^2)

(1) 0.33 m

(2) 3.33 m

(3) 33.3 m

(4) None of these

36) Two blocks of masses 1 kg and 2 kg are connected by a metal wire going over a smooth pulley. The breaking stress of metal is $\frac{40}{3\pi} \times 10^6$ N/m^2. What should be the minimum radius of wire used, if it should not be broken? (g = 10ms^{-2})

(1) 0.5 mm

(2) 1 mm

(3) 1.5 mm

(4) 2 mm

37) If x longitudinal strain is produced in a wire of Young's modulus y, then energy stored in the material of the wire per unit volume is :

(1) yx^2

(2) $2yx^2$

(3) $\frac{1}{2}y^2x$

(4) $\frac{1}{2}yx^2$

38) **Assertion (A) :** Rubber is highly elastic as compared to steel.
Reason (R) : Young's modulus for steel is much larger than that of rubber and so rubber in more elastic as it can be easily stretched to larger lengths and regain its shape.

AIM : FREE EDUCATION TO ALL APUL

(1) Both assertion and reason are correct and assertion is the correct explanation of reason

(2) Both assertion and reason are correct and assertion is not the correct explanation of reason

(3) Assertion is correct but reason is incorrect

(4) Both assertion and reason are incorrect

39) The density of a solid at normal pressure is ρ. When the solid is subjected to an excess pressure p, the density changes to ρ' if the bulk modulus of the solid is K then the ratio ρ'/ρ is :

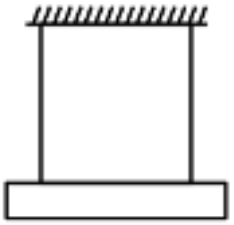
(1) $\frac{p}{K}$

(2) $1 - \frac{p}{K}$

(3) $2 + \frac{p}{K}$

(4) $1 + \frac{p}{K}$

40) Two wires of equal length and cross-section area suspended as shown in figure. Their young's



modulus are Y_1 and Y_2 respectively. The equivalent young's modulus will be:

(1) $Y_1 + Y_2$

(2) $\frac{Y_1 + Y_2}{2}$

(3) $\frac{Y_1 Y_2}{Y_1 + Y_2}$

(4) $\sqrt{Y_1 Y_2}$

41) Two metal spheres are falling through a liquid of density 2×10^3 kg/m^3 with the same uniform speed. The material density of sphere 1 and sphere 2 are 8×10^3 kg/m^3 and 11×10^3 kg/m^3 respectively. The ratio of their radii is :

(1) $\frac{11}{8}$

(2) $\sqrt{\frac{11}{8}}$

(3) $\frac{3}{2}$

(4) $\sqrt{\frac{3}{2}}$

42) In an experiment to verify Stokes law, a small spherical ball of radius r and density ρ falls under gravity through a distance h in air before entering a tank of water. If the terminal velocity of the ball inside water is same as its velocity just before entering the water surface, then the value of h is

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL — APUL

proportional to :
(ignore viscosity of air)

(1) r
(2) r^4
(3) r^3
(4) r^2

43) In the given diagram, there is a liquid film of thickness d between plate and table. On releasing from rest the plate moves towards right with constant velocity v. Find the coefficient of viscosity of liquid.



(1) $\frac{(M_1 - M_2)g}{Av}$
(2) $\frac{M_1 gd}{Av}$
(3) $\frac{M_2 gd}{Av}$
(4) $\frac{(M_1 + M_2)gd}{Av}$

44) The limbs of a U shape tube consist of uniform capillary tubes of diametres 1 mm and 0.5 mm, find out the water level difference in both limbs. [Surface tension = 72×10^{-3} N/m ($\theta = 0°$)]

(1) 28.8 mm
(2) 52.2 mm
(3) 14.4 mm
(4) 25.6 mm

45) An air bubble of radius r in water is at a depth h below the water surface at some instant. If P is atmospheric pressure, d and T are density and surface tension of water respectively, the pressure inside the bubble will be:

(1) P + h dg – (4T/r)
(2) P + h dg + (2T/r)
(3) P + h dg – (2T/r)
(4) P + h dg + (4T/r)

CHEMISTRY

1) Match the list I and list II and pick the correct matching from the code given below :-

List I		List II	
A	Isotopes	a	$^{228}_{88}Ra$ & $^{228}_{89}Ac$
B	Isobars	b	$^{39}_{18}Ar$ & $^{40}_{19}K$
C	Isotones	c	$^{2}_{1}H$ & $^{3}_{1}H$
D	Isosteres	d	$^{235}_{92}U$ & $^{231}_{90}Th$
E	Isodiapheres	e	CO_2 & N_2O

	A	B	C	D	E
(1)	b	a	d	c	c
(2)	b	e	a	d	c
(3)	e	d	a	b	c
(4)	c	a	b	e	d

(1) 1

(2) 2

(3) 3

(4) 4

2) It is known that atoms contain protons, neutrons and electrons. If the mass of the neutron is assumed to be half of its original value whereas that of electron is assumed to be twice of its original value. The atomic mass of $_6C^{12}$ will be :

(1) Twice

(2) 75% less

(3) 25% less

(4) one-half of its original value

3) In Rutherford's experiment most of the α-particles went straight through the gold foil because

(1) α-particles are positively charged

(2) α-particles move with high speed

(3) Most part of the atom is empty space

(4) There is no nucleus in an atom

4) A 100 watt bulb emits light of λ = 400 Å, the number of photons/sec emitted by the bulb will be-

(1) 5×10^{16}

(2) 2×10^{19}

(3) 20×10^{29}

(4) 5×10^{15}

5) If the shortest wavelength of spectral line of H-atom in Lyman series is x, then match the following for Li^{2+}.

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL — APUL

Column-I		Column-II	
(A)	Shortest wavelength in Lyman series	(P)	$\frac{4x}{5}$
(B)	Longest wavelength in Lyman series	(Q)	$\frac{4x}{9}$
(C)	Shortest wavelength in Balmer series	(R)	$\frac{x}{9}$
(D)	Longest wavelength in Balmer series	(S)	$\frac{4x}{27}$

(1) (A)-R, (B)-S, (C)-Q, (D)-P

(2) (A)-Q, (B)-P, (C)-R, (D)-S

(3) (A)-R, (B)-P, (C)-Q, (D)-S

(4) (A)-S, (B)-R, (C)-P, (D)-Q

6) Time taken by an electron to complete one revolution in the Bohr orbit of H-atom is :-

(1) $\frac{4\pi^2 mr^2}{nh}$

(2) $\frac{nh}{4\pi^2 mr^2}$

(3) $\frac{nh}{4\pi mr}$

(4) $\frac{h}{2\pi mr}$

7) The ratio of magnetic moments of Fe^{+3} and Co^{+2} is:-

(1) $\sqrt{5} : \sqrt{7}$

(2) $\sqrt{35} : \sqrt{15}$

(3) 7 : 3

(4) $\sqrt{24} : \sqrt{15}$

8) Calculate the ratio of energy $\frac{E_1 - E_2}{E_2 - E_3}$ for H-atom ?

(1) $\frac{4}{9}$

(2) $\frac{1}{3}$

(3) $\frac{27}{5}$

(4) $\frac{1}{2}$

9) What is the maximum number of electrons that can be associated with d-subshell with n = 5 and m = 0?

(1) 2

(2) 10

(3) 6

(4) 30

10) **Statement-I:** Bohr's theory accounts for the stability and line spectrum of Li^+ ion.
Statement-II: Bohr's theory was unable to explain the splitting of spectral lines in the presence of a magnetic field.

(1) Both statements I and II are true.

(2) Statement I is false but statement II is true.

(3) Both statements I and II are false.

(4) Statement I is true but statement II is false.

11) Number of all possible sub-shell of n + □ = 5 is :-

(1) 2

(2) 3

(3) 4

(4) 5

12) Which of the following order is incorrect about given property?

(1) $HOCl < HClO_2 < HClO_3 < HClO_4$ (Acidic character)

(2) $MnO < MnO_2 < Mn_2O_7$ (Acidic character)

(3) $SiO_2 < Al_2O_3 < MgO < Na_2O$ (Basic Nature)

(4) $F^{\ominus} < Cl^{\ominus} < H^{\ominus} < Br^{\ominus} < I^{\ominus}$ (Ionic radius)

13) Identify the incorrect option ?

(1) ns > np > nd > nf (Penetration power)

(2) $CF_4 > CH_4$ (Electronegativity of 'C')

(3) Li < Na = K = Rb (Screening Constant)

(4) Li < Na < K < Rb > Ca (Metallic character)

14) The increasing order of electron affinity of the electronic configuration of element is ?
(I) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$
(II) $1s^2\ 2s^2\ 2p^3$
(III) $1s^2\ 2s^2\ 2p^5$
(IV) $1s^2\ 2s^2\ 2p^6\ 3s^1$

(1) II < IV < III < I

(2) I < II < III < IV

(3) I < III < II < IV

(4) IV < III < II < I

AIM : FREE EDUCATION TO ALL APUL

15) Process which are endothermic in nature ?
(a) $Ne \rightarrow Ne^{+}$ (b) $F \rightarrow F^{\ominus}$
(c) $O \rightarrow O^{-2}$ (d) $N^{\ominus} \rightarrow N$
(e) $P \rightarrow P^{\ominus}$

(1) a, e
(2) b, c, e
(3) a, c
(4) b, d, e

16) In $IF_6^{\ominus}$ & $TeF_5^{\ominus}$ the sum of axial d-orbitals which are used in hybridisation in both species :-

(1) 5
(2) 4
(3) 3
(4) 2

17) Identify the incorrect match ?

(1)	$Be^{\ominus} + B \rightarrow B^{\ominus} + Be$	Spontaneous Reaction
(2)	+780, +590	II electron gain enthalpy of 'O' & 'S' respectively
(3)	$[Rn]6d^2 7s^2$	d-Block element
(4)	Technetium (Tc)	First man made element

(1) 1
(2) 2
(3) 3
(4) 4

18) Identify the correct order :-
(a) $Cl^{\ominus} > F^{\ominus}$ (I. P.) (b) $O > S^{\ominus}$ (I. P.)
(c) $Cl^{\oplus} < F$ (E. A.) (d) $Cr^{\oplus} < Mn^{\oplus}$ (I. P.)

(1) a, c, d
(2) a, b
(3) b, c, d
(4) a, b, d

19) Which option would be correct?

(1) Zn, Cd, Hg, Cn are part of II B group
(2) As per americans Z = 104 = kurchatovium
(3) As per property $_2$'He' belongs to s-Block
(4) p-Block elements are not part of main group elements.

20) Which of the following statement/option would be incorrect?

AIM : FREE EDUCATION TO ALL APUL

(1) S > Se (ΔH_{eg} order)

(2) HF < HCl (acidic strength)

(3) As, Te → Metals

(4) $Li^{+} > Al^{+3}$ (size)

21) Which of the following is incorrect about given table :-

			C
	B		
D		72	
		A	

(1) Atomic No. of A = 104

(2) Atomic No. of B = 39

(3) Atomic No. of C = 23

(4) Atomic No. of D = 70

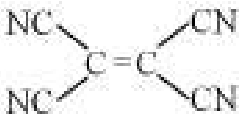
22) How many maximum total elements may be present upto 9th period of periodic table after expansion ?

(1) 50

(2) 168

(3) 218

(4) 290

23) Ratio of total σ bonds and π bonds in given compound is :- 

(1) One

(2) Two

(3) Four

(4) Nine

24) Correct order of metallic Character?

(1) Mg < Si < P < Be < Na

(2) P < Mg < Si < Be < Na

(3) P < Si < Be < Mg < Na

(4) Be < Mg < Si < Na < P

25) Which of the following option would be correct? (as per M.O.T.)

(1) $B_2 > C_2$ (Bond energy)

(2) $O_2 > N_2$ (Bond order)

AIM : FREE EDUCATION TO ALL APUL

(3) $C_2 > O_2$ (Bond length)

(4) All

26) Which option would be not incorrect?

(1) SiF_6^{-2} & $SiCl_6^{-2}$ both exist

(2) CO_3^{-2} has one coordinate bond possible.

(3) Valency of 'Al' in $[Al(H_2O)_5Cl]$ is 6

(4) SO_3 is a hypovalent species

27) Correct order of radius ?

(1) Ti < Zr < Hf

(2) B < Al < Ga < In < T□

(3) Sc < Y < La

(4) O < N < P < S

28) Identify the d-block element

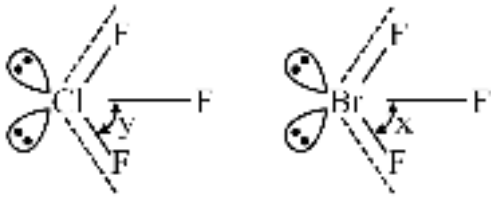
(1) [Xe] $4f^{14}$ $5d^{10}$ $6s^2$ $6p^2$

(2) [Xe] $4f^{14}$ $5d^1$ $6s^2$

(3) [Rn] $5f^{14}$ $6d^1$ $7s^2$

(4) [Xe] $4f^{14}$ $5d^8$ $6s^2$

29) Compare the bond angle from the following molecules and choose correct option?



(1) x > y

(2) y > x

(3) x = y

(4) None

30)

Which option would be incorrect?

(1) SF_4 → Folded square shape

(2) MgO > NaCl Lattice energy

(3) $Cu^+ > K^+$ Polarising Power

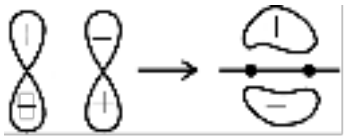
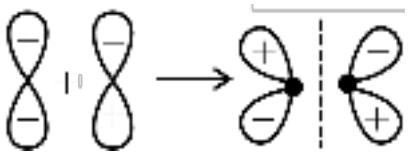
(4) XeF_2 → Linear shape, diagonal hybridisation

AIM : FREE EDUCATION TO ALL APUL

31) **Assertion :** Reactivity of alkali metals increases down the group while that of halogens it decreases down the group.
Reason : In a group of alkali metals & halogen e^- gain enthalpy regulary decreases down the group respectively.

(1) Both Assertion & Reason are true & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are true but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but the Reason is False.
(4) Both Assertion & Reason are False.

32) Which of the following linear combinations of atomic orbitals is incorrectly depicted ?

(1)  → σ*2pz
(2)  → σ2pz
(3)  → π 2px
(4)  → π* 2px

33) Match List-I with List-II

	List-I (Molecule / Ion)		**List-II (Type of bonds / Overlapping)**
(A)	$NO_3^{\ominus}$	(i)	One pπ-pπ bond, Two dπ-pπ bond
(B)	SO_3	(ii)	σ ⇒ sp-p overlapping
(C)	$BeCl_2$	(iii)	σ ⇒ sp^3-s overlapping
(D)	CH_4	(iv)	One pπ-pπ bond only

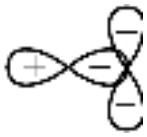
(1) (A) - (ii), (B) - (iv), (C) - (i), (D) - (iii)
(2) (A) - (iv), (B) - (i), (C) - (ii), (D) - (iii)
(3) (A) - (iv), (B) - (i), (C) - (iii), (D) - (ii)
(4) (A) - (iii), (B) - (iv), (C) - (i), (D) - (ii)

34) Match List-I & List-II

	List-I		**List-II**
(A)		(i)	Negative overlap
(B)		(ii)	In phase overlap

AIM : FREE EDUCATION TO ALL ♥ APUL

AIM : FREE EDUCATION TO ALL APUL

(C)		(iii)	σ-bond
(D)		(iv)	π-bond
		(v)	Zero overlap

(1) A-(ii, iii), B-(i), C-(v), D-(ii, iv)

(2) A-(i, iii), B-(ii, iii), C-(v), D-(i, iv)

(3) A-(i, iii), B-(i, iii), C-(iv, v), D-(i)

(4) A-(ii, iii), B-(i), C-(iv), D-(ii, v)

35) Incorrect match is-

	Hybridisation	**Geometry**	**Orbitals use**
(1)	sp^3d	Trigonal Bipyramidal	$s+px+py+pz+dz^2$
(2)	sp^3d^3	Pentagonal Bipyramidal	$s+px+py+pz+dx^2–y^2+dz^2+dxy$
(3)	sp^3d^2	Capped octahedral	$s+px+py+pz+dx^2–y^2+dz^2$
(4)	sp^3	Tetrahedral	$s+px+py+pz$

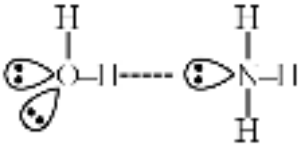
(1) 1

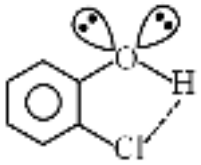
(2) 2

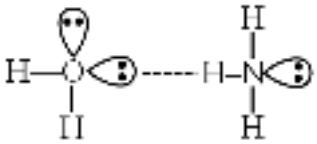
(3) 3

(4) 4

36) Which of the following is not a best representation of H-Bond?

(1) 

(2) 

(3) 

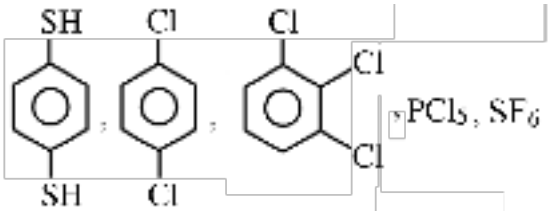
(4) .....H–F.....H–F.....

AIM : FREE EDUCATION TO ALL ❤ APUL

37) Which property is due to H-bonding ?
(a) High boiling point of water
(b) Solubility of NH_3 in Water
(c) Polar nature of Halogen acids
(d) High viscosity of H_3PO_4

(1) a, b & d
(2) a, b & c
(3) a & b
(4) a, b, c & d

38) Number of species with zero dipole moment are :-

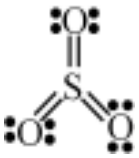
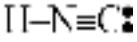
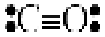
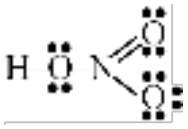


(1) 2
(2) 3
(3) 4
(4) 5

39) If 2s-2p mixing is not considered then magnetic nature and bond order of B_2 :-

(1) Paramagnetic, 2
(2) Diamagnetic, 2
(3) Paramagnetic, 1
(4) Diamagnetic, 1

40) Which of the following Lewis structure has zero formal charge on each atom ?

(1) 
(2) 
(3) 
(4) 

41) Which of the following order of bond length is correct ?

(1) O – H < C – N < C – C < C ≡ N
(2) O – H < C ≡ N < C – N < C – C
(3) C ≡ N < O – H < C – N < C – C

(4) $C \equiv N < C - N < C - C < O - H$

42) Hybridisation of 'Xe' in XeF_6(s) & 'P' in PBr_5(s) respectively is :-

(1) sp^2 & sp^3d
(2) sp^3d^2 & sp^3
(3) sp^3d^2 & sp^2
(4) sp^2 & sp^3

43) Correct order of bond angle?

(1) $NO_2^{\ominus} > NO_3^{\ominus} > NO_2^{\oplus}$
(2) $BF_3 > BCl_3 > BBr_3 > BI_3$
(3) $CO_3^{-2} = NO_3^{\ominus} < NO_2^{\oplus}$
(4) $SOF_2 > SOCl_2 > SOBr_2$ $(X - \hat{S} - X)$

44) Correct order according to given property is :-

(1) He > Ne < Ar < Kr > Xe (Ease of liquification)
(2) $CO_2 < CS_2 < CCl_4 > H_2O$ (Boiling point)
(3) ICl > Br_2 (Boiling point)
(4) $H_2O > H_2O_2$ (Boiling point)

45) Ionisation potential of Be is higher than :-
(a) B (b) Mg (c) Li (d) Al (e) K

(1) a, b, c only
(2) a, b, d only
(3) a, b, c, d only
(4) a, b, c, d, e

BIOLOGY

1) Find out the correct ascending order of molecules in a cell according to percentage composition :-

(1) Ions < Lipids < Carbohydrates < Nucleic Acids < Proteins
(2) Lipids < Ions < Carbohydrates < Nucleic Acids < Proteins
(3) Proteins < Nucleic Acids < Carbohydrates < Lipids < Ions
(4) Proteins < Carbohydrates < Nucleic Acids < Lipids < Ions

2) Which one of the following is the correct increasing order of occurance of elements by % weight in human body ?

(1) O < C < H < N
(2) Ca < S < Na < Mg

AIM : FREE EDUCATION TO ALL APUL

(3) Ca < N < C < O

(4) N < Ca < S < O

3) Which acid is represent by
$CH_3–(CH_2)_{14}$ –COOH

(1) Palmitic Acid

(2) Stearic Acid

(3) Arachidonic Acid

(4) Oleic Acid

4) **Statement I :** The manner in which subunits of protein are arranged with respect to each other is known quartnery structure of a protein
Statement II : Two subunits of α-type and two subunits of β-type together constitute the human haemoglobin .

(1) Both statements are correct

(2) Statement I is correct and II is incorrect

(3) Statement I is incorrect and II is correct

(4) Both statements are incorrect

5) Two statements A & B are given. Choose the correct answer related to the statements.
Statement A :- Amino acids are amphoteric in nature
Statement B :- Except glycine all amino acids have two enantiomeric isomers.

(1) Statement A is wrong, statement B is correct.

(2) Both the statement A & B are wrong

(3) Both the statement A & B are correct

(4) Statement A is correct, Statement B is wrong

6) Identify the correct match from the columns I, II and III :-

Column-I		**Column-II**		**Column-III**	
(I)	Phospholipid	(a)	Abrin	(i)	Animal starch
(II)	Hexose	(b)	Glycogen	(ii)	Choline
(III)	Secondary metabolite	(c)	Fructose	(iii)	Toxins
(IV)	Polysaccharide	(d)	Lecithin	(iv)	Fruit sugar

(1) I-d-ii, II-a-iv, III-b-iii, IV-c-i

(2) I-c-iii, II-b-ii, III-a-i, IV-d-iv

(3) I-d-ii, II-c-iv, III-a-iii, IV-b-i

(4) I-a-ii, II-b-iii, III-c-i, IV-d-iv

7)

AIM : FREE EDUCATION TO ALL APUL

Find out the correct match between column I and column II.

Column-I		Column-II	
(a)	GLUT – 4	**(i)**	Most abundant protein in the biosphere
(b)	Collagen	**(ii)**	Transporter protein
(c)	Insulin	**(iii)**	Enzyme
(d)	Trypsin	**(iv)**	Intercellular ground substance
(e)	RuBisCO	**(v)**	Hormone

(1) a–v, b–iv, c–ii, d–i, e–iii

(2) a–ii, b–iv, c–v, d–iii, e–i

(3) a–iv, b–ii, c–v, d–i, e–iii

(4) a–ii, b–iv, c–iii, d–v, e–i

8) **Assertion :-** Coenzymes are loosely bound cofactor.
Reason :- Their association with the apoenzyme is only transient, usually occuring during the course of catalysis.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

9) Read the following four statements (A–D) :-
(A) Thermal stability is an important quality of enzymes which are isolated from thermophilic organisms
(B) Formation of Enzyme substrate complex, during enzymatic reaction, is a facultative phenomenon
(C) Enzymes bring down the energy barrier between substrate and transition state to make the transition of substrate to product easy
(D) The essential chemical components of many coenzymes are proteins and metal ions
Which of the above statements are **correct** ?

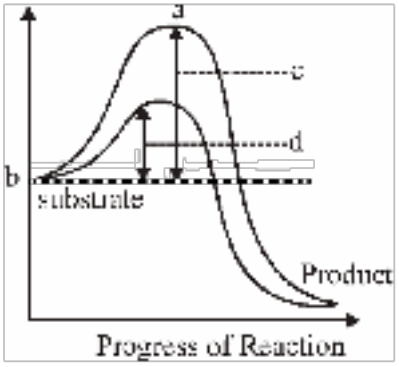
(1) A, B, C & D

(2) Only A

(3) A & C

(4) A, B & C

10) The figure given shows the conversion of substrate into product by an enzyme. In which one of the four options, the components of the reaction labelled a, b, c & d are identified correctly?



(i) Potential energy

AIM : FREE EDUCATION TO ALL APUL

(ii) Transition state
(iii) Activation energy with enzyme
(iv) Activation energy without enzyme

(1) a-i b-ii c-iii d-iv
(2) a-ii b-i c-iv d-iii
(3) a-ii b-i c-iii d-iv
(4) a-i b-ii c-iv d-iii

11) The lowest molecular weight compounds in acid insoluble fraction are :-

(1) Proteins
(2) Nucleic acids
(3) Polysaccharides
(4) Lipids

12) Protein part of a conjugate enzyme is known as :-

(1) Apoenzyme
(2) Coenzyme
(3) Prosthetic group
(4) Cofactor

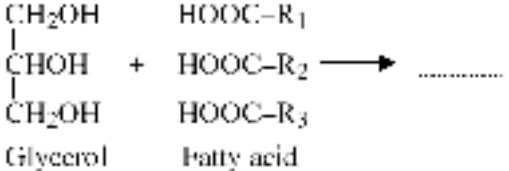
13) Cellulose is the most important constituent of plant cell wall is made up of :-

(1) Branched chain of glucose molecule by α 1, 6 linkage
(2) Unbranched chain of glucose molecule by β 1, 4 linkage
(3) Unbranched chain of glucose molecule by α 1, 4 linkage
(4) Branched chain of glucose molecule by β 1, 6 linkage

14) In competitive inhibition :-

(1) Inhibitor resembles the substrate in molecular structure
(2) There is competition between substrate and inhibitor to occupy active site
(3) Binding of the inhibitor to active site declines the enzyme action
(4) All of these

15) The product of the given reaction would be a



(1) Monoglyceride
(2) Diglyceride
(3) Triglyceride
(4) Both(1) & (3)

AIM : FREE EDUCATION TO ALL APUL

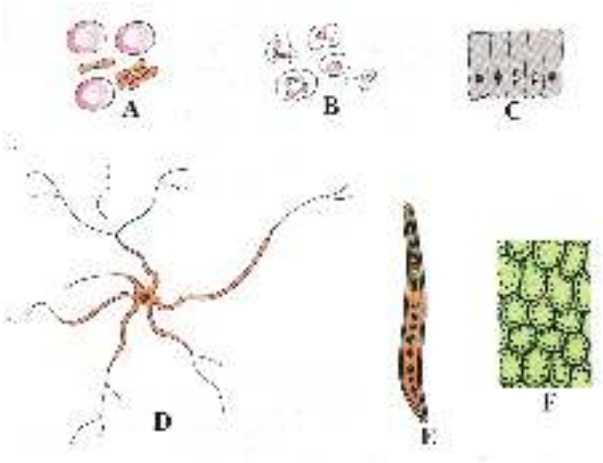
16) (A) Unicellular organisms are capable of independent existence
(B) Cell is the fundamental, structural and functional unit of all living organisms
(C) All cells arise from pre-existing cells
(D) The cytoplasm is the main arena of cellular activities in eukaryotic cell only.
Option :-

(1) Statement A,C and D are wrong
(2) Statement A, B and D are not correct
(3) Statement A, B and C are not wrong
(4) Statement B, C and D are not correct

17) The correct order of size among following :-

(1) Eukaryotic cell > PPLO > Viruses > Bacteria
(2) Eukaryotic cell > Bacteria > PPLO > Viruses
(3) Eukaryotic cell > PPLO > Bacteria > Viruses
(4) Eukaryotic cell > Viruses > PPLO > Bacteria

18) Given below is the diagram showing different types of cells. Identify the cells labelled A, B, C, D, E, F and select the right option about them.



	A	**B**	**C**	**D**	**E**	**F**
(1)	Round and oval	Amoeboid	Long and narrow	Branched and long	Round and biconcave	Elongated
(2)	Round and oval	Amoeboid	Branched and long	Long and narrow	Elongated	Round and biconcave
(3)	Round and biconcave	Amoeboid	Branched and long	Long and narrow	Elongated	Round & biconcave
(4)	Round and biconcave	Amoeboid	Long and narrow	Branched and long	Elongated	Round and oval

(1) 1
(2) 2

AIM : FREE EDUCATION TO ALL APUL

(3) 3

(4) 4

19) Read the following four statements (a-d) :-
(a) The quasifluid nature of lipid enables lateral movement of protein with in the overall bilayer.
(b) Membrane of R.B.C. has approximately 52% protein and 40% lipid
(c) Neutral solutes may move across the membrane by the process of simple diffusion.
(d) Peripheral proteins are partially or totally burried in the membrane.
How many of the above statements are correct?

(1) Three

(2) Four

(3) Two

(4) One

20) (a) Primary cell wall is capable of growth.
(b) A non-living rigid structure called as cell wall, forms an outer covering for the plasma membrane of fungi and plants.
(c) Secondary cell wall gradually diminishes as the cell matures and the primary cell wall is formed.
(d) The cell wall and middle lamellae may be traversed by plasmodesmata.
Choose the correct option for correct statements:-

(1) a, c and d

(2) a, b, c and d

(3) b, c and d

(4) a, b and d

21) Which of the following statements are true about Endoplasmic reticulum?
A – Smooth Endoplasmic Reticulum makes lipids.
B – It is also called the control centre of cell.
C – It synthesises the nucleic acid.
D – It often shows ribosomes attached to their outer surface.
Options :

(1) A, B, C and D are incorrect

(2) A and B are correct

(3) B, C and D are incorrect

(4) A and D are correct

22) **Statement-1:** A cell organelle with distinct cis and trans faces, remains in close association with endoplasmic reticulum in all living cells.
Statement-2: This cell organelle with cis and trans faces synthesise many glycoproteins which are packaged and secreted by endoplasmic reticulum.
Above out the two statements :-

(1) Statement - 1 and statement - 2 both are correct

(2) Statement - 1 and statement - 2 both are incorrect

(3) Statement - 1 is correct but statement - 2 is incorrect

AIM : FREE EDUCATION TO ALL ❤ APUL

(4) Statement - 1 is incorrect but statement - 2 is correct

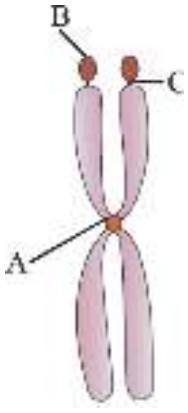
23) Which of the following statement is **not** correct ?

(1) Lysosomes have numerous hydrolytic enzymes.
(2) The hydrolytic enzymes of lysosomes are active under alkaline pH.
(3) Lysosomes are membrane bound structures.
(4) Lysosomes are formed by the process of packaging in the golgi body.

24) From the column-I, column-II, column-III how many of the following are correctly match –

	Column-I	**Column-II**	**Column-III**
(A)	Ribosome	Composed of r-RNA and protein	Site of protein synthesis
(B)	ER	Network of tiny tubular structure scattered in the cytoplasm	Site for synthesis for steroidal hormone
(C)	Kinetochore	Disc shape structure present on secondary constriction of chromosome	Site of attachment for spindle fibres
(D)	Microbody	Membrane bound minute vesicle	Present only in plant cell

(1) 3
(2) 2
(3) 1
(4) 4



25) What does A, B and C represent in the given figure of a chromosome?

	A	**B**	**C**
1	Centriole	Satellite	Primary constriction
2	Centriole	Satellite	Secondary constriction

AIM : FREE EDUCATION TO ALL APUL

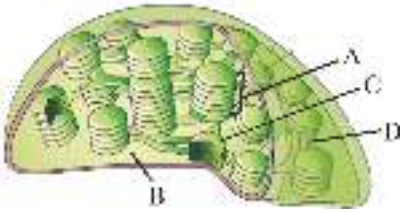
3	Centromere	Satellite	Secondary constriction
4	Centromere	Satellite	Primary constriction

(1) 1

(2) 2

(3) 3

(4) 4



26) Recognise the figure and find out the correct matching :-

(1) A–Stroma, B–Granum, C–Thylakoid, D–Stroma lamella

(2) A–Granum, B–Stroma, C–Thylakoid, D–Stroma lamella

(3) A–Thylakoid, B–Stroma, C–Stroma lamella, D–Granum

(4) A–Thylakoid, B–Stroma, C–Granum, D–Stroma lamella

27) Match Column-I with Column-II :-

Column-I		**Column- II**	
(I)	Histone protein	(A)	Lightly stained
(II)	Euchromatin	(B)	Membrane less
(III)	Heterochromatin	(C)	Basic amino acid
(IV)	Nucleolus	(D)	Genetically less active or inert

(1) I-A, II-B, III-C, IV-D

(2) I-C, II-A, III-B, IV-D

(3) I-C, II-A, III-D, IV-B

(4) I-C, II-B, III-D, IV-A

28) **Assertion :-** The cell membrane is selectively permeable.
Reason :- Many molecules can move across the membrane without any requirement of energy.

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true and reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Both assertion and reason are false.

29) **Assertion :** In plant cells, chloroplasts are responsible for trapping light energy essential for photosynthesis.
Reason : The stroma in the plastid, is the site of light reaction and grana of dark reaction.

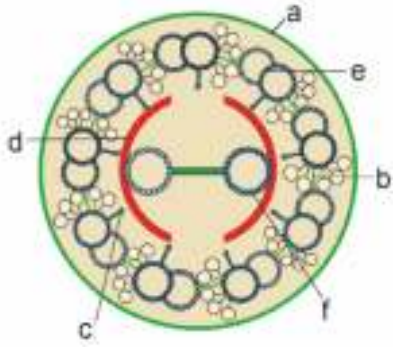
AIM : FREE EDUCATION TO ALL APUL

(1) Both assertion and reason are correct and reason is the correct explanation of assertion.

(2) Both assertion and reason are correct but reason is not the correct explanation of assertion.

(3) Assertion is correct but reason is incorrect.

(4) Assertion is incorrect but reasons correct.



30) Recognise the figure and find out the correct matching.

(1) d–plasma membrane, b–radial spoke, a–central sheath, c–interdoublet bridge, f–peripheral microtubule, e–central microtubule

(2) d–plasma membrane, c–radial spoke, a–central sheath, b–interdoublet bridge, e–peripheral microtubule, f–central microtubule

(3) a–plasma membrane, b–radial spoke, d–central sheath, c–interdoublet bridge, e–peripheral microtubule, f–central microtubule

(4) a–plasma membrane, c–radial spoke, d–central sheath, f–central microtubule.

31) **Assertion:** The centrioles form the basal body of cilia or flagella, and spindle fibres that give rise to spindle apparatus during cell division.
Reason: Centriole are made up of nine unevenly spaced peripheral fibrils of tubulin protein in animal cells.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

32) **Assertion (A) :** The content of nucleolus is continuous with the rest of nucleoplasm.
Reason (R) : Nucleolus is not a membrane bound structure.

(1) Assertion is not correct but Reason is correct.

(2) Both Assertion and Reason are correct and the Reason is the correct explanation of Assertion.

(3) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.

(4) Assertion is correct but Reason is not correct.

33) **Assertion :** The organisation of the prokaryotic cell is fundamentally similar even though prokaryotes exhibit a wide variety of shapes and functions.
Reason : All prokaryotes have a cell wall surrounding the cell membrane except in mycoplasma.

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true but reason is not correct explanation of assertion.

AIM : FREE EDUCATION TO ALL APUL

(3) Assertion is true but reason is false.

(4) Assertion is false but reason is true.

34) If a cell has 12 bivalents in zygotene stage how many kinetochores are found in it ?

(1) 24

(2) 48

(3) 12

(4) 36

35) The diagram below shows the nuclear contents of a cell.  Which line in the following table refers correctly to this cell ?

	Number of pairs of homologous chromosomes	Number of chromatids	Number of centromeres
(1)	3	6	12
(2)	3	12	6
(3)	6	6	12
(4)	3	12	9

(1) 1

(2) 2

(3) 3

(4) 4

36) Match the events of prophase-I given in column-I with substages of prophase-I given in column-II and select the correct option from options given below :

	Column-I (Events of prophase-I)		Column-II (Sub stage of prophase-I)
(A)	Initiation of condensation of chromatin	(P)	Pachytene
(B)	Chiasmata formation	(Q)	Diakinesis
(C)	Crossing over	(R)	Laptotene
(D)	Terminalisation of chiasmata	(S)	Diplotene

AIM : FREE EDUCATION TO ALL APUL

(E)	Pairing of homologous chromosome	(T)	Zygotene

	A	B	C	D	E
(1)	T	S	P	Q	R
(2)	P	Q	R	S	T
(3)	S	R	P	Q	T
(4)	R	S	P	Q	T

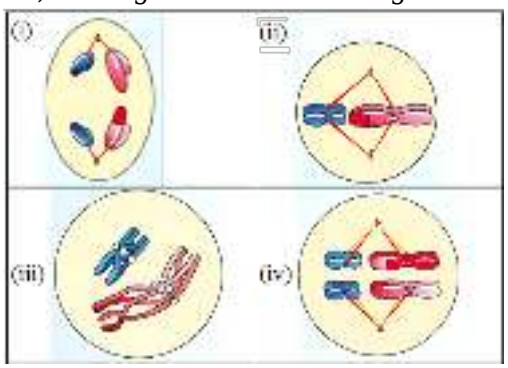
(1) 1

(2) 2

(3) 3

(4) 4

37) Arrange the different stages of meiosis (i), (ii), (iii) and (iv) in sequence :



(1) (i) → (iv) → (ii) → (iii)

(2) (iii) → (iv) → (ii) → (i)

(3) (iv) → (iii) → (ii) → (i)

(4) (ii) → (i) → (iii) → (iv)

38) Which of the following are not characteristics of anaphase-I?
(A) Centromere split and chromatids separate.
(B) Chromosome move to opposite poles.
(C) Homologous chromosomes separate.
(D) Nuclear membrane and nucleolus reappear.
(E) Bivalent chromosomes align on the equatorial plate.

(1) A, D and E

(2) B and C

(3) C, D and E

(4) C and D

39) Mark incorrect statement :

(1) Each centrosome radiates out microtubules called asters

(2) The two asters together with spindle fibres form mitotic-apparatus

(3) Prophase follows the G_0 and S phases of interphase

(4) Prophase is the first stage of karyokinesis

AIM : FREE EDUCATION TO ALL ♥ APUL

40) **Statement-I:** Tubulin and some other proteins are synthesised for preparation of mitosis in G_2 phase.
Statement-II: During G_1 phase the cell is metabolically active and replicate their DNA.

(1) Statement I and statement II both are correct
(2) Statement I is correct but II is incorrect
(3) Statement I is incorrect but II is correct
(4) Statement I and statement II both are incorrect

41) Match List-I with List-II

	List-I		**List-II**
A.	Synapsis	I	Sites where crossing over occurred
B.	Chiasmata	II	Similar in shape, size & genetically similar
C.	Bivalent	III	Pairing of two homologous chromosome
D.	Homologous chromosome	IV	Complex formed by a pair of synapsed homologous chromosomes

Choose the correct answer :
(1) A-III, B-I, C-IV, D-II
(2) A-II, B-I, C-III, D-IV
(3) A-III, B-I, C-II, D-IV
(4) A-II, B-III, C-IV, D-I

42) In which phase of meiosis, the homologous chromosomes separate, while sister chromatids remains associated at their centromeres ?

(1) Anaphase I
(2) Anaphase II
(3) Metaphase I
(4) Telophase I

43) Identify the stage with correct description.

(1) Diakinesis - The compaction of chromosomes is not completed in the stage.
(2) Diplotene- The beginning of this stage is recognised by the dissolution of the synaptonemal complex.
(3) Leptotene-Chromosomes start pairing together and this process is called synapsis
(4) Pachytene-It represents transition to metaphase.

44) **Statement-I :** In cell cycle at the time of cytoplasmic division, organelles like mitochondria and plastids get distributed between two daughter cells.
Statement-II : In animals cytokinesis occurs by constriction and furrow formation.

(1) Only statement I is correct

AIM : FREE EDUCATION TO ALL APUL

(2) Only statement II is correct

(3) Both statement are correct

(4) Both statement are incorrect

45) **Statement I :-** M phase involving a major reorganisation of virtually all components of the cell.
Statement II :- The number of chromosomes in the parent and progeny cells is the same, it is called as equational division.

(1) Both Statement I & II are correct.

(2) Both Statement I & II are incorrect.

(3) Statement I is correct and II is incorrect.

(4) Statement II is correct and I is incorrect.

46) Match following column-I with column-II & III.

	Column-I		**Column-II**		**Column-III**
A.	Vital capacity	i.	IRV + TV + ERV + RV	P.	3500 ml
B.	Inspiratory capacity	ii.	45 mm Hg	Q.	4500 ml
C.	pCO_2 in tissue	iii.	IRV + TV	R.	High pCO_2
D.	Total lung capacity	iv.	IRV + ERV + TV	S.	6000 ml

(1) A-iv-Q, B-iii-R, C-i-S, D-ii-P

(2) A-ii-Q, B-iv-S, C-i-R, D-ii-P

(3) A-iv-Q, B-iii-P, C-ii-R, D-i-S

(4) A-iv-Q, B-iii-P, C-ii-S, D-i-R

47) **Assertion :** Relaxation of the diaphragm and the external Intercostal muscle Reduce the thoracic volume and there by the pulmonary volume.
Reason : Relaxation of diaphragm and external inter costal muscle causes an increases in intra-pulmonary pressure to slightly above the atmospheric pressure causing the expulsion of air from the lungs.

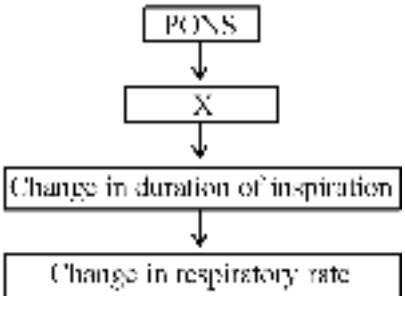
(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

48) Study the given flow chart about the regulation of respiration



In the above flow chart 'X' is :-

AIM : FREE EDUCATION TO ALL APUL

(1) Medulla

(2) Respiratory rhythm center

(3) Pneumotaxic center

(4) Chemosensitive area

49) **Statement-I :-** Nearly 20-25% of CO_2 is transported as in the form of bicarbonate in plasma.
Statement-II :- The role of O_2 in the regulation of respiration is quite significant.
Choose the appropriate option :-

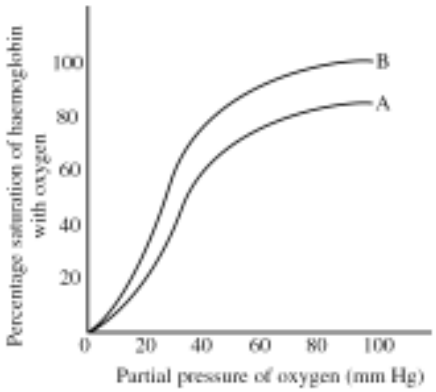
(1) Statement-I and statement-II both are correct.

(2) Statement-I and statement-II both are incorrect.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

50) Given below is the oxygen dissociation curve. Choose correct condition from below in which this curve will shift from A to B.



(1) Increase in partial pressure of CO_2

(2) Decrease in pH

(3) Increase in pH

(4) Increase H^+ ion & temperature

51) **Assertion :** Emphysema leads to poor oxygenation of the blood.
Reason : The alveolar surface area for gaseous exchange is reduced in emphysema.

(1) Both **Assertion** and **Reason** are false

(2) **Assertion** is true but **Reason** is false.

(3) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion.**

(4) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

52) Select the incorrect statement form following.

(1) 3% of O_2 is carried in a dissolved sate through the plasma.

(2) The total thickness of diffusion membrane is much less than a nanometre.

(3) A healthy man can inspire or expire approximately 6000 to 8000 ml of air per minute.

(4) Pharynx is a common passage for food and water.

AIM : FREE EDUCATION TO ALL APUL

53) Match list I with II

List-I		List-II	
A.	pO_2 in atmospheric air	(i)	95 mm Hg
B.	pCO_2 in alveoli	(ii)	104 mm Hg
C.	pO_2 in alveoli	(iii)	40 mm Hg
D.	pO_2 in oxygenated blood	(iv)	159 mm Hg

Choose the correct answer from the options given below :

(1) A-(ii), B-(ii), C-(iv), D-(i)

(2) A-(iv), B-(iii), C-(ii), D-(i)

(3) A-(i), B-(ii), C-(iv), D-(iii)

(4) A-(i), B-(iii), C-(iv), D-(ii)

54) **Assertion :-** The power of self contraction and relaxation is not present in mammalian lungs.
Reason :- Mammalian lungs are solid, spongy and without muscles.

(1) Assertion and reason both are true and the reason is not correct explaination of assertion.

(2) Assertion and reason both are true and the reason is correct explaination of assertion.

(3) Assertion is true but the reason is false.

(4) Assertion and reason both are false.

55) Which of the following animal has cutaneous and pulmonary both types of respiration

(1) Earthworm

(2) Frog

(3) Prawn

(4) Cockroach

56) C-shaped cartilagenous rings in Human respiratory tract are found ?

(1) From trachea upto end of bronchioles

(2) From trachea upto alveoli

(3) Only in trachea and primary bronchi

(4) From trachea to initial bronchioles

57) Binding of oxygen with haemoglobin is related to which factor ?

(1) PO_2

(2) PCO_2

(3) H^+ ion concentration

(4) All of the above

58) Diffusion membrane for exchange of gases is formed by how many major layers -

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

(1) 5
(2) 3
(3) 2
(4) 6

59) Nasopharynx open through _____ of the larynx region into the trachea.

(1) glottis
(2) gullet
(3) epiglottis
(4) pharynx

60) Read following terminology of column-I & II and identify correct match :-

	Column-I		**Column-II**
(A)	Systolic pressure	(i)	Electrocardiogram
(B)	Stroke volume	(ii)	5000 ml/min
(C)	Calcium	(iii)	70 ml/min
(D)	End of T wave	(iv)	End of ventricular systole
		(v)	Helps is blood clotting
		(vi)	120 mm of Hg
		(vii)	Prevent clotting

(1) A-(i), B-(iv), C-(ii), D-(vi)
(2) A-(i), B-(ii), C-(iv), D-(v)
(3) A-(ii), B-(iii), C-(iv), D-(v)
(4) A-(vi), B-(iii), C-(v), D-(iv)

61) Read the column-I, II and III and identify their correct match -

Column-I		**Column-II**		**Column-III**	
(A)	Fishes	(P)	3-chambered heart	(i)	Single circulation
(B)	Amphibia	(Q)	4-chambered heart	(ii)	Pale yellow in colour
(C)	Mammals	(R)	2-chambered heart	(iii)	Double circulation
(D)	Plasma	(S)	Matrix of blood	(iv)	Incomplete double circulation

(1) (A)-(R)-(iv), (B)-(P)-(iii), (C)-(Q)-(iv), (D)-(S)-(iii)
(2) (A)-(R)-(i), (B)-(Q)-(iv), (C)-(P)-(iv), (D)-(S)-(ii)
(3) (A)-(R)-(i), (B)-(P)-(iv), (C)-(Q)-(iii), (D)-(S)-(ii)
(4) (A)-(R)-(iii), (B)-(P)-(iv), (C)-(Q)-(iii), (D)-(S)-(ii)

62) Read the following statements (A-D)
(A) To obtain a standard ECG, a patient is connected to the machine with three electrical leads (One

to each wrist and to the left ankle) that continuously monitor the heart activity
(B) By counting in the number of QRS complexes that occur in a given time period, on can determine the heart beat rate of an individual
(C) ECG obtained from different individuals have roughly the same shape for a given lead configuration
(D) The P-wave represents the electrical excitation (or depolarisation) of the atria, which leads to the contraction of both the atria
How many of the above statements are true :-

(1) 3
(2) 4
(3) 1
(4) 2

63) Which of the following is correct ?

(1) Glucose, amino acids and lipids are present in the plasma as they are always in transit in the body.
(2) Platelets are not included under formed elements.
(3) A healthy individual has 12-16 mg of Hb/100 mL of blood.
(4) All the blood cells are nucleated.

64) **Statement-I :** Ventricular systole increases the pressure causing the closure of AV valves due to attempted back flow of blood into the atria.
Statement-II : Ventricular diastole cases the closure of SL valves which prevents the backflow of blood into the ventricles.

(1) Both Statement-I and Statement-II are incorrect.
(2) Statement-I is incorrect but Statement-II is correct.
(3) Both Statement-I and Statement-II are correct
(4) Statement-I is correct but Statement-II is incorrect

65) Which one of the following is **not correct** ?

(1) Normal activities of heart are regulated by specialised muscles tissues.
(2) Parasympathetic neural signals decreases the heart rate.
(3) Adrenal medullary hormones can also decreases the cardiac output.
(4) Human heart is myogenic & autoexcitable

66) **Assertion :** Blood group O^- individuals are universal donors.
Reason : Blood group O^- individual RBCs lack all type of antigens (A, B & Rh)

(1) Both Assertion and Reason are correct and Reason is not the correct explanation of Assertion.
(2) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
(3) Assertion is correct but Reason is incorrect
(4) Assertion is incorrect but Reason is correct.

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL ❤ APUL

67) Match the Column-I with Column-II-

	Column-I		**Column-II**
(a)	Basophils	(i)	Immune response
(b)	Globulin	(ii)	2%
(c)	Lymphocytes	(iii)	Osmotic balance
(d)	Albumin	(iv)	Release histamine

Select the correct option-

	(a)	**(b)**	**(c)**	**(d)**
(1)	iv	ii	i	iii
(2)	iv	i	ii	iii
(3)	iii	i	ii	iv
(4)	ii	iii	i	iv

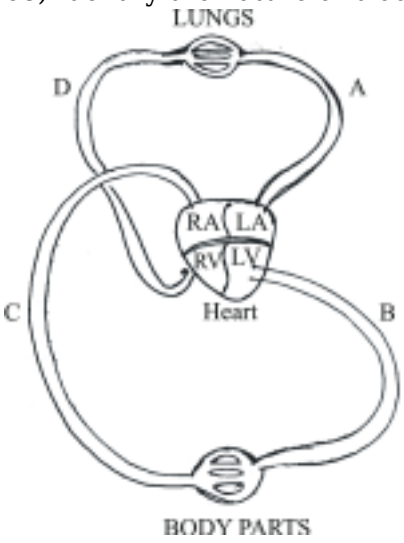
(1) 1

(2) 2

(3) 3

(4) 4

68) Identify the nature of blood passing through blood vessels A, B, C and D ?



(1) A and B deoxygenated, C and D oxygenated

(2) A and B oxygenated, C and D deoxygenated

(3) A and C oxygenated, B and D deoxygenated

(4) B and C oxygenated, A and D deoxygenated

69) Why body fluids and their circulation are required in animals -

(1) To provide nutrients, O_2 and other essential substances to the cells of body.

(2) For the continuously removal of harmful waste from body.

(3) Lymph and plasma do not have similar mineral distribution.
(4) (1) and (2) both

70) Average life span of R.B.Cs is

(1) 80 days
(2) Few hours
(3) 120 days
(4) 24 hours

71) Which of the following is not required for clotting of blood ?

(1) Ca^{2+}
(2) Prothrombin
(3) Thrombokinase
(4) Heparin

72) In human heart the opening between the right atrium and the right ventricle is guarded by which type of valve.

(1) Bicuspid valve
(2) Thebasian valve
(3) Eustachian valve
(4) Tricuspid valve

73) Which component of the conduction system present at lower left corner of right atria?

(1) SA node
(2) AV node
(3) Left bundle branch
(4) Purkinje fibres

74) If the cardiac output of a person is 6400 ml and the heart rate of his person is 64 /min then what should be the stroke volume of this person :-

(1) 70 ml
(2) 100 ml
(3) 640 ml
(4) 96 ml

75) Fluid called lymph is almost similar to blood except for -

(1) Protein content
(2) Mineral content
(3) Formed elements
(4) 1 & 3 both

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL

76) In Hypertension which organ of our body is unaffected ?

(1) Kidney
(2) Brain
(3) Heart
(4) Bones

77) Maximum surface area of circulatory system is covered by :-

(1) Arteries
(2) Veins
(3) Capillaries
(4) Heart

78) If vagus nerve is cut the heart beat

(1) increases
(2) decreases
(3) will not be affected
(4) will stop immediately

79) Tubular fluid is ___(A)____ to plasma as it enters Bomman's capsule, ___(B)___ to plasma at beginning of LOH, ___(C)___ to plasma at the base of loop in medulla, ___(D)___ to plasma as it leaves the LOH to enter the DCT and become ___(E)___ to plasma when leaves the collecting duct (Respectively)

(1) Isotonic, Hypertonic, Hypertonic, Isotonic, Hypertonic
(2) Isotonic, Isotonic, Hypertonic, Hypotonic, Hypertonic
(3) Isotonic, Isotonic, Hypotonic, Hypertonic, Hypotonic
(4) Isotonic, Hypotonic, Hypotonic, Isotonic, Hypotonic

80) How many of the following statements are true?
Statement 1 : Blood is pumped into a dialyzing unit, where it passes through a coiled cellophane tube surrounded by a fluid (dialyzing fluid) with the same composition as plasma, except for nitrogenous wastes.
Statement 2 : The porous cellophane membrane allows the removal of urea and other substances from the blood in the dialyzing unit.
Statement 3 : The composition of the dialyzing fluid is similar to that of plasma, excluding nitrogenous wastes.
Statement 4 : An anti-coagulant like heparin is added to the blood before it is returned to the body during the process of hemodialysis.

(1) 1
(2) 2
(3) 4
(4) 3

AIM : FREE EDUCATION TO ALL APUL

81) Read the following statements and choose the correct option
Statement A : An excessive loss of fluid from the body can activate osmoreceptors which stimulate the hypothalamus to secrete antidiuretic hormone.
Statement B : Counter current mechanism helps to maintain a concentration gradient in the medullary interstitium.

(1) Both the statement are correct
(2) Both the statement are Incorrect
(3) Only statement A is correct
(4) Only statement B is correct

82) **Statement-I :** In Juxta medullary nephrons, loop of Henle is very short and runs deep into the medulla.
Statement-II : Vasa recta is absent or highly reduce in cortical nephrons.

(1) Both statement are correct
(2) Statement I is correct only.
(3) Statement II is correct only.
(4) Both statement are incorrect.

83) Match column I with column II

Column I		**Column II**	
(i)	Renal failure	(a)	inflammation of glomerulus.
(ii)	Diabets Mellitus	(b)	Stone or isoluble mass of crystalised salt
(iii)	Glomerulonephritis	(c)	Presence of glucose & ketone bodies in urine
(iv)	Renal calculi	(d)	Malfunctioning of kidney

(1) i→a,ii→b,iii→c,iv→d
(2) i→c,ii→b,iii→d,iv→a
(3) i→d,ii→c,iii→a,iv→b
(4) i→a,ii→b,iii→c,iv→d

84) Which of the following statement is incorrect ?

(1) The epithelial cells of Bowman's capsule called podocytes are arranged in an intricate manner so as to leave some minute spaces called filtration slits or slit pores.
(2) Blood is filtered so finely through filtration membrane, that almost all the constituents of the plasma except the proteins pass onto the lumen of the Bowman's capsule.
(3) The glomerular capillary blood pressure causes filtration of blood through 3 layers, i.e., the endothelium of glomerular blood vessels, the epithelium of Bowman's capsule and a basement membrane between these two layers.

(4) A decrease in blood flow to the atria of the heart can cause the release of Atrial Natriuretic Factor.

85) **Assertion (A) :** Tubular secretion is an important step in urine formation.
Reason (R) : During urine formation tubular cells secrete like H^+, K^+ and a ammonia to maintain ionic and acid base balance of body fluids.

(1) Both (A) & (R) are true and (R) is not correct explanation of (A)
(2) (A) is true but (R) is false
(3) (A) is false but (R) is true
(4) Both (A) & (R) are true and (R) is correct explanation of (A)

86) Which of the following substance may be retained in the kidney matrix of some of the animals to maintain a desired osmolarity?

(1) Uric acid
(2) Ammonia
(3) Urea
(4) CO_2

87) The cortex extends in between the medullary pyramids as renal columns called

(1) Hilum
(2) Calyces
(3) Columns of Bertini
(4) Capsule

88) Liver eliminates all of the following except :-

(1) biliverdin
(2) Lactic acid
(3) Vitamins
(4) degraded steroid hormones

89) Urination/micturition occurs when :

(1) Urinary bladder relaxes and urethral sphincter contracts
(2) Urinary bladder and urethral sphincter both contracts
(3) Urinary bladder contracts and urethral sphincter relaxes
(4) Urinary bladder and urethral sphincter both relaxes

90) A network of minute blood vessel runs parallel to the Henle's loop called

(1) Vasa recta
(2) Afferent arteriole
(3) Efferent arteriole

AIM : FREE EDUCATION TO ALL APUL

(4) Renal artery

AIM : FREE EDUCATION TO ALL ❤ APUL

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	3	1	1	3	1	1	1	3	4	4	2	3	1	4	2	2	4	3	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	4	4	4	2	4	2	2	3	2	3	3	2	4	3	2	4	4	4	2
Q.	41	42	43	44	45															
A.	4	2	2	1	2															

CHEMISTRY

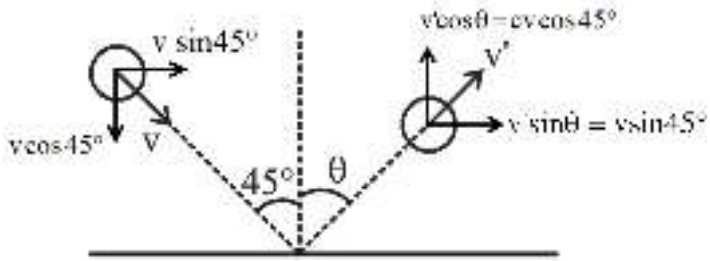
Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	3	3	2	1	1	2	3	1	2	2	4	3	1	3	2	3	2	1	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	3	1	3	3	3	3	4	2	4	3	1	2	1	3	3	1	2	4	1
Q.	86	87	88	89	90															
A.	2	2	3	3	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	3	1	1	3	3	2	1	3	2	4	1	2	4	3	3	2	4	1	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	4	2	2	2	3	2	3	2	3	4	3	2	2	2	2	4	2	1	3	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	1	2	3	1	3	1	3	2	3	3	2	2	2	2	4	4	2	1	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	2	1	3	3	2	1	2	4	3	4	4	2	2	4	4	3	1	2	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	3	3	4	4	3	3	2	3	1										

AIM : FREE EDUCATION TO ALL ♥ APUL

PHYSICS



1)

During collision between ball and surface velocity in horizontal direction remain same.

$v \sin 45° = v' \sin\theta$ ...(1)

$ev \cos 45° = v' \cos\theta$ ...(2)

Equation (1) ÷ Equation (2)

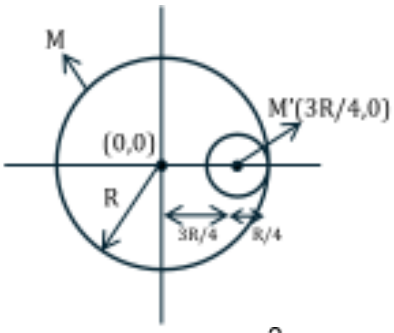
$\frac{1}{e}\tan 45° = \tan\theta$ $\quad \therefore \tan 45° = 1; \quad \therefore e = \frac{1}{\sqrt{3}}$

$\frac{1}{\frac{1}{\sqrt{3}}} = \tan\theta = \sqrt{3}$

$\therefore \theta = 60º$

2) no change in momentum along horizontal direction.

3)



$$M' = \frac{-M}{\pi R^2} \times \pi\left(\frac{R}{4}\right)^2 = \frac{M}{16}$$

$$X_{com} = \frac{m_1X_1 + m_2x_2}{m_1 + m_2}$$

$$X_{com} = \frac{\frac{M \times O - \frac{M}{16} \times \frac{3R}{4}}{4}}{M - \frac{M}{16}}$$

$$= \frac{\frac{-3R}{64}}{15}16 = \frac{-R}{20}$$

(–ve sin used for direction Left side)

4)

AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL ❤ APUL

By using conservation of momentum

$\vec{P_i} = \vec{P_f}$

$0 = m\vec{v}_1 + m\vec{v}_2$

$\therefore\ \vec{v}_1 = -\vec{v}_2$

$u_1 = 6$ m/s ; $u_2 = -6$ m/s

$$\frac{1}{3} = \frac{V_2 - (-V_2)}{6 - (-6)} = \frac{2V_2}{12}$$

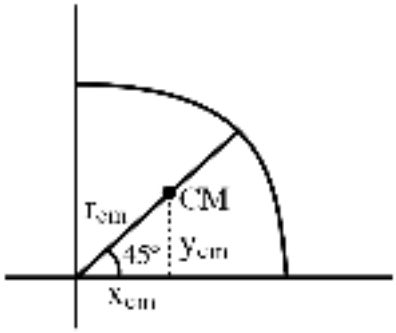
$V_2 = 2$ m/s

$V_1 = -V_2 = -2$ m/s

5)

$x_{cm} = r_{cm}\cos 45°$

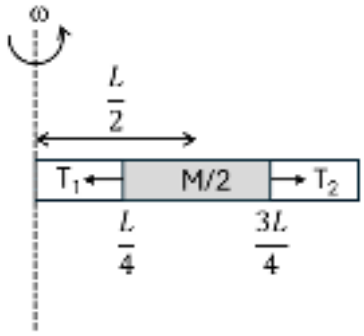
$y_{cm} = r_{cm}\sin 45°$



$$r_{cm} = \frac{2R\sin 45°}{3(\pi/4)}$$

$$r_{cm} = \frac{4\sqrt{2}R}{3\pi}$$

6)



Centripetal force $F_c = m\omega^2 r$

$$T_1 - T_2 = \frac{M}{2}(\omega^2)\frac{L}{2} \quad \Rightarrow \quad T_1 > T_2$$

7) **Solution:**

COME : $K_1 + U_1 = K_2 + U_2$

$0 + mg(4R) = \frac{1}{2}mv^2 + mg(2R) \Rightarrow mv^2 = 4mgR$

Forces at position 2 :

AIM : FREE EDUCATION TO ALL APUL

$N = \frac{mv^2}{R} - mg = 4mg - mg = 3\,mg$

8) **Solution :**

The necessary centripetal force $\left(F_C = \frac{mv^2}{R}\right)$ required during a turn is supplied by the friction between the tyres of vehicle and the road .

For safe turning, the centripetal force must be less than limiting friction (f_L), i.e. $F_C \leq f_L$.

$\left(\frac{mv2}{R} \leqslant \mu mg\right)$

Since friction directly provides the centripetal force, the reason correctly explains the assertion.

9)

$\theta = (4t^3 - 3t^2 + 2t + 1)$

$\omega = \frac{d\theta}{dt} = 12t^2 - 6t + 2,$

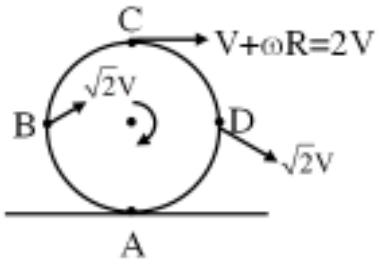
at t = 2 sec, $\omega = 12 \times (2)^2 - 6 \times 2 + 2 = 38$ rad/sec

$\alpha = \frac{d\omega}{dt} = 24t - 6,$

at t = 2 sec, $\alpha = 24 \times 2 - 6 = 42$ rad/sec^2

10)

Solution :-



v = 4 m/s

R = 2m

v = ωR

ω = 2 rad/sec

$v_B = v_D = \sqrt{2}v = 4\sqrt{2}$

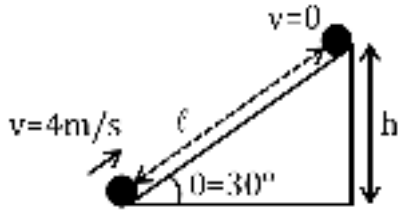
$v_A = 0$

$v_C = 2v = 8$ m/s

11)

Solution :

The initial kinetic energy of the rolling cylinder (which is the sum of its translational and rotational kinetic energy) is converted entirely into gravitational potential energy at its highest point on the incline.



$$\frac{1}{2}mv^2\left(1+\frac{K^2}{R^2}\right)=mgh$$

$$\Rightarrow 8\left(1+\frac{1}{2}\right)=10h \Rightarrow h = 1.2\ m$$

$$\frac{h}{\ell}=\sin 30^\circ \Rightarrow \ell = 2.4\ m$$

12) **Solution :**

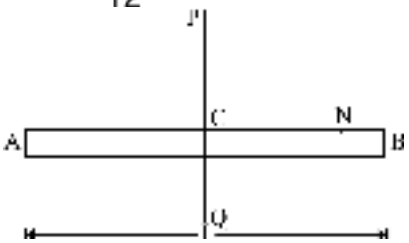
$$I = 0+0+m\left(\frac{\ell}{\sqrt{2}}\right)^2+m\left(\frac{\ell}{\sqrt{2}}\right)^2$$

$= m\ell^2$

13)

Moment of inertia of the rod about a perpendicular axis PQ passing through the centre of the mass C,

$$I_{CM}=\frac{ML^2}{12}$$



Let N be the point which divides the length of the rod AB in ratio 1 : 3. This point will be at a distance $\frac{L}{6}$ from C. Thus, the moment of inertia I' about an axis parallel to PQ and passing through the point N.

$$I' = I_{CM}+M\left(\frac{L}{6}\right)^2=\frac{ML^2}{12}+\frac{ML^2}{36}=\frac{ML^2}{9}$$

If K be the radius of gyration, then

$$K=\sqrt{\frac{I'}{M}}=\sqrt{\frac{L^2}{9}}=\frac{L}{3}$$

14)

$N_1 + N_2 = mg$

$$N_1=\frac{mg}{4}$$

$$N_2=\frac{3mg}{4}$$

$\tau_{ACW} = \tau_{CW}$

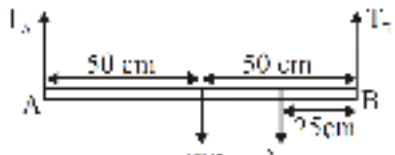
AIM : FREE EDUCATION TO ALL ❤ APUL

$N_1 \times \frac{L}{2} = N_2 \times x$

$\frac{mg}{4} \times \frac{L}{2} = \frac{3mg}{4} \times x$

$x = \frac{L}{6}$

Distance $\frac{L}{2} - \frac{L}{6} = \frac{L}{3}$



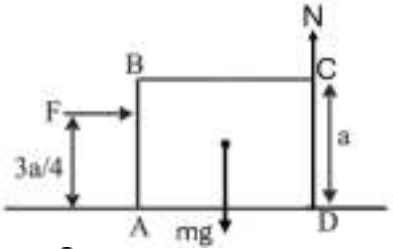
15) $\tau B = 0$ (torque about point B is zero)

$(T_A) \times 100 - (mg) \times 50 - (2mg) \times 25 = 0$

$100\ T_A = 100\ mg$

$T_A = 1\ mg$

16) **Solution:** Then the block will topple down, when torque due to F balances torque due to weight.



$F \times \frac{3a}{4} = mg \times \frac{a}{2}$

$\Rightarrow F = \frac{2}{3}mg$

17) **Solution :**

$\omega_0 = 1200\ \text{rpm} = \frac{1200}{60} = 20\ \text{rev/s}$

$\Rightarrow \omega_0 = 20 \cdot 2\pi = 40\pi\ \text{rad/s}$

Given $\alpha = -2\ \text{rad/s}^2$, and final speed $\omega = 0$:

$\theta = \frac{\omega_0^2}{2\alpha} = \frac{(40\pi)^2}{2 \cdot 2} = \frac{1600\pi^2}{4} = 400\pi^2\ \text{radians}$

$n = \frac{\theta}{2\pi} = \frac{400\pi^2}{2\pi} = 200\pi \approx 628\ \text{revolutions}$

18) Work $= \frac{1}{2}I\omega^2 = \frac{1}{2}(1)\frac{(40 \times 10^{-2})^2}{2}(10 \times 2\pi)^2$

$= \frac{1}{4}(16)\ (10^{-2})100\ (4\pi^2) = 16\pi^2 = 158\ J$

19)

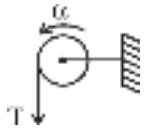
AIM : FREE EDUCATION TO ALL APUL

AIM : FREE EDUCATION TO ALL APUL



FBD of Block

$mg - T = ma$ ..(1)



FBD of Disc

$T.R = I.\alpha \Rightarrow T.R = I.\frac{a}{R}$

$T = I\frac{a}{R^2}$ ..(2)

from eq^n (1) & (2)

$mg - I\frac{a}{R^2} = ma \Rightarrow a = \frac{mg}{m + \frac{I}{R^2}}$

Hence acceleration of the block

$= \frac{mg}{m + \frac{I}{R^2}} = \frac{mg}{m + \frac{m}{2}} = \frac{2}{3}g$

so velocity $v = \sqrt{2ah} = \sqrt{\frac{4gh}{3}} = 2\sqrt{\frac{gh}{3}}$

Alternate solution :

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{mR^2}{2}\right)\left(\frac{V^2}{R^2}\right)$$

$$V = 2\sqrt{\frac{gh}{3}}$$

20) $v_A = v_B$

$r_A \omega_A = r_B \omega_B$

$r_A \omega_A = 3r_A\omega_B \quad \Rightarrow \omega_A = 3\omega_B$

Now, $L_A = L_B$

$I_A\omega_A = I_B\omega_B$

$I_A(3\omega_B) = I_B\omega_B$

$\frac{I_A}{I_B} = \frac{1}{3}$

21)

Gravitational potential at centre of solid sphere

$V_{center} = \frac{-3}{2}\frac{GM}{R}$

Escape velocity from earth surface is given by

$V_e = \sqrt{\frac{2GM}{R}} \approx 11.2 km/sec$

Kepler's second law $\frac{dA}{dt}$ = constant

Total energy of satellite is given by

$T.E = \frac{-GMm}{2(R+h)}$, if $h \approx 0$

AIM : FREE EDUCATION TO ALL APUL

then $T.E = \frac{-GMm}{2R}$

22)

Energy of satellite in orbit of radius r = $-\frac{GMm}{2r}$

Energy of satellite in orbit of radius $\frac{3}{2}r = -\frac{GMm}{2 \times \frac{3}{2}r}$

Increased in energy = $\frac{-GMm}{2 \times \frac{3}{2}r} - \left(-\frac{GMm}{2r}\right)$

$\Delta E = -\frac{GMm}{3r} + \frac{GMm}{2r} = \frac{GMm}{6r}$

23) $\frac{GM}{R_e^2} \times \frac{1}{100} = \frac{GM}{(R_e + h)^2}$

$(R_e + h)^2 = 100R_e^2$

$h = 9R_e$

24) w = mg = 100 N

w' = mg'

g' = $g_d = g\left(1 - \frac{d}{R}\right)$

= $g\left(1 - \frac{R}{4R}\right) = \frac{3g}{4}$

$w' = \frac{3mg}{4} = \frac{3}{4} \times 100 = 75\,N$

25) $g = \frac{GM}{R^2}$

$\frac{\Delta g}{g} \times 100 = \frac{\Delta M}{M} \times 100 - \frac{2\Delta R}{R} \times 100$

= (–1%) –2(2%)

= –5%

26) $L = mvr = m\left(\sqrt{\frac{GM}{r}}\right) r = m\sqrt{GMr}$

$\therefore L \propto \sqrt{r}$

27)

From COME :-

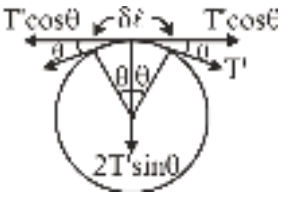
$-\frac{GMm}{R} + \frac{1}{2}m\left(\frac{1}{3}\sqrt{\frac{2GM}{R}}\right)^2 = -\frac{GMm}{R+h}$

$$= \frac{-8}{9R} = -\frac{1}{R+h}$$

$$h = \frac{R}{8}$$

28) $K.E. = \frac{GMm}{2r}$ K.E. $\propto r^{-1} T^2 \propto r^3$ $r \propto T^{2/3}$

$\therefore$ K.E. $\propto T^{-2/3}$

29) 

$\Rightarrow$ $2T'\sin\theta = Tdl$

$2T'\theta = T(2\theta.R)$

Tension $T' = RT$]

30)

$P_1V_1 = P_2V_2$

$\Rightarrow (P^0 + \rho gH) \times \frac{4}{3}\pi r^3 = P^0 \times \frac{4}{3}\pi(2r)^3$

$\Rightarrow P_0 + \rho gH = 8P_0$

$\Rightarrow \rho gH = 7P_0$

$\Rightarrow H = 70$ m

31) For pressure we consider only vertical distances means height, so pressure at piston = P

$$P = \frac{F}{A} = (\rho_2 h_2 + \rho_3 h_3)g$$

32) Level of water in the vessel will be maximum when rate of inflow of water = rate of outflow of water

or $\alpha = av$

or $\alpha = a\sqrt{2gh}$

or $h = \frac{\alpha^2}{2ga^2}$

i.e., $h \propto \alpha^2$ and $h \propto a^{-2}$.

33) $\nu = \sqrt{2gh}, t = \sqrt{\frac{2h}{g}}$

and $R = 2\sqrt{h(H-h)}$

At mountain, value of g will be less. Hence, ν will decreases, t will increases and R will remain unchanged.

AIM : FREE EDUCATION TO ALL APUL

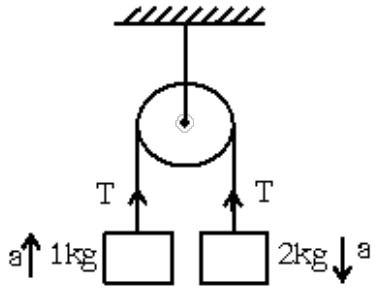
AIM : FREE EDUCATION TO ALL ❤ APUL

34) Normal stress = $\frac{F_{\perp ar}}{A^1} = \frac{F\cos\theta}{A}$ $\cos\theta$ $\left[A^1 = \frac{A}{\cos\theta}\right]$

35) $\text{Stress} = \frac{\text{weight}}{\text{area}} = \frac{mg}{A} = \frac{Vdg}{A} = \frac{ALdg}{A} = Ldg$

$\therefore L = \frac{\text{breaking stress}}{dg} = \frac{10^6}{3 \times 10^3 \times 10} = 33.3\,\text{m}.$

36) Here T – g = a ----------(1)
2g – T = 2a ----------(2)



⇒ 2g – T = 2 (T – g)

⇒ $\boxed{T = \frac{4g}{3}}$

For minimum radius r :- $\frac{T}{\pi r^2}$ = Stress

⇒ $r = \sqrt{\frac{T}{\pi \times \text{stress}}}$

⇒ $r = \sqrt{\frac{4g}{3\pi \times \text{stress}}} = \sqrt{\frac{40 \times 3\pi}{3\pi \times 40 \times 10^6}} = \frac{1}{1000}\text{m} = 1\text{mm}$

37) Stress = y × strain = xy. So energy stored per unit volume = $\frac{1}{2}x^2y$

38) Both assertion and reason are false as elasticity is proportional to the resistance to deformation within the range of Hook's law or elastic limit. Young's modulus is more for steel as it undergoes lesser strain for same load as compared to rubber. Rubber is more stretchable but for less elastic. Term elasticity should not be confused with amount of stretching.

39) $\frac{\Delta V}{V} = \frac{\frac{M}{\rho'} - \frac{M}{\rho}}{\frac{M}{\rho}} = \frac{\rho - \rho'}{\rho'}$

Bulk modulus K = $\frac{\frac{-p}{\Delta V}}{V} \Rightarrow \frac{\Delta V}{V} = \frac{-p}{K}$

$\frac{\rho'}{\rho} = \left(1 - \frac{p}{K}\right)^{-1} = 1 + \frac{p}{K}$

AIM : FREE EDUCATION TO ALL APUL

40)

$F = F_1 + F_2$

$$\frac{(2A)Y\Delta L}{L} = \frac{AY_1\Delta L}{L} + \frac{AY_2\Delta L}{L}$$

$2Y = Y_1 + Y_2$

$$Y = \frac{Y_1 + Y_2}{2}$$

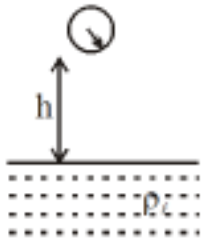
41)

$$V = \frac{2}{9\eta} r^2 (\rho - \sigma)g$$

Here $v_1 = v_2$

$$r_1^2(\rho_1 - \sigma) = r_2^2(\rho_2 - \sigma)$$

$$\frac{r_1^2}{r_2} = \frac{11-2}{8-2} = \frac{3}{2} \Rightarrow \frac{r_1}{r_2} = \sqrt{\frac{3}{2}}$$

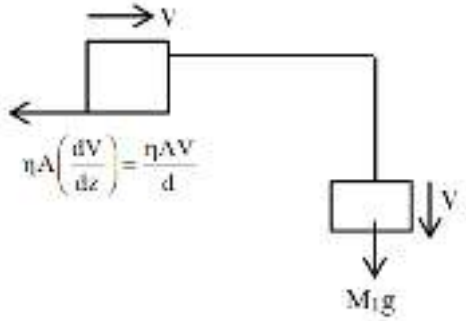


42)

After falling through h, the velocity be equal to terminal velocity

$$\sqrt{2gh} = \frac{2}{9}\frac{r^2 g}{\eta}(\rho - \rho_w)$$

$$\Rightarrow \boxed{h \propto r^4}$$



43)

$$\frac{\eta Av}{d} = M_1 g$$

$$\eta = \frac{M_1 gd}{Av}$$

44) $h_1 - h_2 = \dfrac{2T\cos\theta}{\rho g}\left\{\dfrac{1}{r_1} - \dfrac{1}{r_2}\right\}$

$$= \frac{2 \times 72 \times 10^{-3} \times 1}{10^3 \times 10} \left\{ \frac{1}{\frac{0.5 \times 10^{-3}}{2}} - \frac{1}{\frac{10^{-3}}{2}} \right\}$$

= 28.8 mm

45) Pressure at a depth h inside water or pressure outside the bubble = P + hdg

As excess pressure inside the air bubble $= \frac{2T}{r}$.

 Pressure inside the air bubble $= P + hdg + \frac{2T}{r}$.

CHEMISTRY

46)

A. Isotopes (A → c) : Atoms with same atomic number (Z) but different mass number (A).
Example : ^{12}H and ^{13}H (same Z = 1, different A = 2, 3).
B. Isobars (B → a) : Atoms with same mass number (A) but different atomic number (Z).
Example : ^{228}Ra and ^{228}Ac (A = 228, Z = 88 & 89).
C. Isotones (C → b) : Atoms with same number of neutrons (N = A–Z).
Example : ^{39}Ar and ^{40}K (N = 21 in both).
D. Isosteres (D → e) : Molecules with same number of atoms and same number of valence electrons.
Example : CO_2 and N_2O (3 atoms, 16 valence electrons each).
E. Isodiapheres (E → d) : Atoms with same neutron-proton difference (N–Z).
Example : ^{235}U and ^{231}Th (N–Z = 51 for both)

47) $m_N = \frac{mN}{2} = 3amu$

New

$m_e = 2\ m_e$

New

Atomic mass (New) = $m_P + m_N$

= 6 + 3 = 9 amu

So, % = $\frac{12-9}{12} \times 100 = 25\%$

48)

Most part of atom empty
So, α-particles (mostly) went straight through gold foil.

49)

$E = \frac{nRC}{\lambda}$

AIM : FREE EDUCATION TO ALL APUL

$$n = \frac{E\lambda}{RC}$$
$n = 2 \times 10^{19}$

50)

Rydberg equation: $1/\lambda = R \cdot Z^2 \cdot (1/n_1^2 - 1/n_2^2)$.
For H (Z = 1), for shortest wavelength of Lyman series ($n_2 = \infty$, $n_1 = 1$) ; $1/x = R \Rightarrow R = 1/x$.
Now for Li^{2+} (Z = 3):

for shortest lyman line $n_2 = \infty$, $n_1 = 1$,
$1/\lambda = R \cdot Z^2 \cdot (1/1^2 - 1/\infty^2) = 9/x$, $\lambda = (x/9) \rightarrow$ **R**

For Longest Lyman ($2 \rightarrow 1$) ,
$1/\lambda = R \cdot Z^2 \cdot (1/1^2 - 1/2^2) = 27/4x$,

So, $\lambda = 4x/27 \rightarrow$ **S**.
For Shortest Balmer ($\infty \rightarrow 2$) ,
$1/\lambda = R \cdot Z^2 \cdot (1/2^2 - 1/\infty^2) = 9/4x$

So , $\lambda = 4x/9 \rightarrow$ **Q**.
For Longest Balmer ($3 \rightarrow 2$) ,
$1/\lambda = R \cdot Z^2 \cdot (1/2^2 - 1/3^2) = 5/4x$

So , $\lambda = 4x/5 \rightarrow$ **P**.

(A)–R, (B)–S, (C)–Q, (D)–P

51) According to Bohr

$$mvr = \frac{nh}{2\pi}$$
$$v = \frac{nh}{2\pi mr}$$
Time = distance/speed
$$= \frac{2\pi r}{\left(\frac{nh}{2\pi mr}\right)} = \frac{4\pi^2 mr^2}{nh}$$

52) $$\frac{Fe^{+3}}{Co^{+2}} = \frac{\sqrt{5(5+2)}}{\sqrt{3(3+2)}} = \frac{\sqrt{35}}{\sqrt{15}}$$

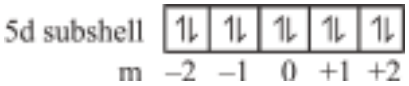
53) $$E = \frac{E_1}{n^2} \quad \Rightarrow \quad \frac{E_1 - E_{1/2^2}}{E_{1/2^2} - E_{1/3^2}} = \frac{27}{5}$$

54)

For a given subshell (d $\rightarrow$ l = 2), the magnetic quantum number (m) can have 5 values: −2, −1, 0, +1, +2.

AIM : FREE EDUCATION TO ALL ❤ APUL

Each **m** value represents **one orbital**, and each orbital can hold **2 electrons** (with spins +½ and −½).



Since m = 0 corresponds to only one orbital, it can have 2 electrons regardless of n.

55)

Bohr's theory applies only to single-electron species (e.g., H, He^{+}, Li^{2+}) ⇒ not valid for Li^{+} (2 electrons) ⇒ False
Bohr's model can't explain spectral splitting in magnetic field (Zeeman effect) ⇒ True
Statement I is false, Statement II is true.

56)

Condition: **n + ℓ = 5**

n = 5, ℓ = 0 → 5s

n = 4, ℓ = 1 → 4p

n = 3, ℓ = 2 → 3d
Hence total subshells = 3

57)

Ionic radius order ⇒
$F^{\ominus} < Cl^{\ominus} < Br^{\ominus} < H^{\ominus} < I^{\ominus}$

58)

σ ⇒ Li < Na < K < Rb < Cs
Z_{eff} = Li < Na = K = Rb = Cs

59)

EA ⇒ $\boxed{N < Na < F < C\ell}$
↓ ↓
$2p^3$ Maximum
Half filled stable EA in P. table
EA ≈ 0

60)

Ne → Ne^{+}

AIM : FREE EDUCATION TO ALL APUL

$O \rightarrow O^{-2}$
These are endothermic process

61)

$IF_6^{\ominus} \Rightarrow$ Hybridisation - sp^3d^3
$TeF_5^{\ominus} \Rightarrow$ Hybridisation - sp^3d^2
Sum of axial d- orbital
$\Rightarrow 2 + 2 = 4$
Axial d-orbitals $\Rightarrow d_{x^2-y^2}, d_{z^2}$

62)

[Rn] $6d^2 7s^2 \Rightarrow {}_{90}Th$
$\Rightarrow$ f-block element (III-B group)

63) $X(g) \underset{IP}{\overset{EA}{\rightleftharpoons}} X^{\ominus}$
[EA of X = IP of X°]
$\therefore$ IP = $Cl^{\circ} > F^{\circ}$
IP = O > S > S°
EA = $S^{\oplus}$ > S > O

64)

NCERT-XI, Part-I, Chapter - 3

65)

As, Te $\rightarrow$ Metals

66)

			C
	B		
D		72	
		A	

A = 72 + 32 = 104
B = 72 – 32 – 1 $\Rightarrow$ 39
C = 23
D = 56

67) $n_e = \frac{(n+1)^2}{2}$ (n = odd)

AIM : FREE EDUCATION TO ALL APUL

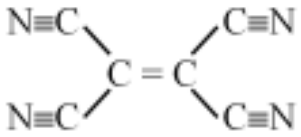
$n_e = \frac{(n+2)^2}{2}$ (n = even)

 Total elements upto 9th Period

⇒ 118 + 50 + 50 ⇒ 218

↓ ↓ ↓

upto 7th pd. 8th pd. 9th pd.

68) 

Total σ = 9

Total π = 9

$\frac{\sigma}{\pi}$ ratio ⇒ $\frac{9}{9} = 1$

69)

Metallic character :

Na > Mg > Be > Si > P

70)

B.O. ∝ B.E. ∝ $\frac{1}{\text{B.L.}}$

71)

C.no. of Al = 6

72)

Atomic radius :

Ti < Zr ≈ Hf

B < Ga ≤ Al < In ≤ T

Sc < Y < La

O < N < S < P

73)

[Xe] $4f^{14}$ $5d^8$ $6s^2$

⇒ d-block element

⇒ Z = 78 - Pt

74)

Bond angle ⇒

AIM : FREE EDUCATION TO ALL APUL

ClF_3 > BrF_3

⟶

EN of C.A. ↓
BA ↓
∴ BA = y > x

75)

LE ∝ $|q_1q_2|$
PP ⇒ [Pseudo noble gas > noble gas]

76)

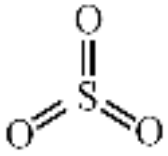
Concept

77)

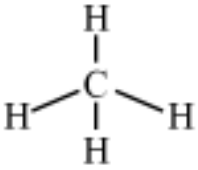
Concept

78)

$NO_3^{\ominus}$ ⇒ Only one pπ-pπ bond

SO_3 ⇒ 

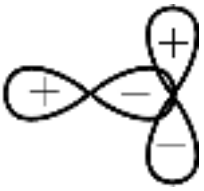
⇒ 1 pπ-pπ bond ; 2 pπ-dπ bond
$BeCl_2$ ⇒ Cl – Be –Cl sp-p σ bond



CH_4 ⇒ sp^3-s σ bond

79)  ⇒ In phase overlap, σ bond

 ⇒ –ve overlap

 ⇒ Zero overlap

AIM : FREE EDUCATION TO ALL ❤ APUL

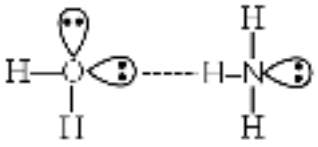
 ⇒ π-bond, in phase overlap

80) 

Capped octahedral

Hyb. – sp^3d^3

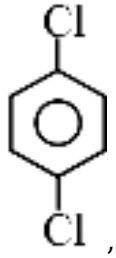
$s + px + py + pz + dx^2 - y^2 + dz^2 + dxy$

81) 

'H' of less EN elements & □p of more EN element wrong selection.

82)

Boiling point, solubility & viscosity are affected by H-Bonding.

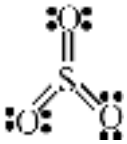
83)  , PCl_5, SF_6 are Non-polar species

84)

If sp mixing Not allowed :

$B_2 = 10e^- \sigma1s^2\ \sigma^*1s^2\ \sigma2s^2\ \sigma^*2s^2\ \underbrace{\sigma^2 2p^2}_{\text{Homo}}$

 Diamagnetic

85) 

AIM : FREE EDUCATION TO ALL ❤ APUL

FC on S, O, O, O → Zero

86)

NCERT Data Table -4.2

87)

$XeF_6(s)$
↓
$XeF_5^+ + F^-$
⇓
sp^3d^2
$PBr_5(s) = PBr_4^+ + Br^-$
sp^3

88)

BA ⇒ CO_3^{-2} = NO_3^- < NO_2^+
–120° –120° –sp Hyb.
– Resonance –Resonance –180°
– sp^2 Hyb –sp^2 –Linear
– T. planar –T. planar

89)

Boiling point → IC > Br_2
- Polar - Non polar
- Keesome - L.D.F
Attraction Attraction

IP → Be > B
Be > Mg
Be > Li
Be > Al
Be > K

90)
a, b, c, d, e Ans.

BIOLOGY

91) NCERT (XI) Pg. # 147

AIM : FREE EDUCATION TO ALL APUL

92)

NCERT-XI, Pg. # 105

93)

NCERT-XI, Pg. # 107, Fig 9.1

94)

NCERT-XI, Pg. # 112

95)

NCERT-XI, Pg. # 106, 107

96)

NCERT XI Pg # 146

97)

NCERT-XI, Pg. # 109, Table 9.5

98) NCERT XI Pg. # 159

99) NCERT-XI Eng. Pg. # 154 (1st para), 156 (1st and 2nd para) and 159 (3rd last para)
NCERT-XI हिन्दी पेज # 154 (दूसरा पेराग्राफ), 156 (दूसरा एवं तीसरा पेराग्राफ) एवं 159 (आखिरी पेराग्राफ)

100)

NCERT-XI, Pg. # 115

101)

NCERT-XI, Pg. # 109

102)

NCERT-XI, Pg. # 117, 118

103)

NCERT XI Pg. # 149, Para No. # 9.5

AIM : FREE EDUCATION TO ALL ❤ APUL

104)

NCERT-XI, Pg. # 117

105)

NCERT-XI, Pg. # 107, Table 9.1

106) NCERT (XI) Pg. # 126, Para–2

107)

NCERT-XI, Pg.# 89

108)

NCERT-XI, Pg. # 89, Fig 8.1

109)

NCERT-XI, Pg. # 93

110)

NCERT-XI, Pg. # 94

111)

NCERT-XI, Pg. # 95, 96

112)

NCERT-XI, Pg. # 95, 96

113)

NCERT-XI, Pg. # 96

114)

NCERT-XI, Pg. # 95, 98, 101, 102

115)

NCERT-XI, Pg. # 101, Fig. 8.13

AIM : FREE EDUCATION TO ALL ❤ APUL

AIM : FREE EDUCATION TO ALL ❤ APUL

116) NCERT (XI) Pg. # 136

117)

NCERT-XI, Pg. # 101

118)

NCERT-XI, Pg. # 94

119)

NCERT-XI, Pg. # 98

120)

NCERT-XI, Pg. # 99

121)

NCERT-XI, Pg. # 99, 100

122) NCERT-XI, Pg. NO. # 100

123)

NCERT-XI, Pg. # 89, 90

124)

The correct answer is:
2.48
Explanation:

A. In the zygotene stage of meiosis I, homologous chromosomes pair up to form bivalents.
B. Each chromosome has 2 chromatids, and each chromatid has 1 kinetochore.
C. Since there are 12 bivalents, this means there are 24 chromosomes (12 pairs).

Each chromosome has 2 kinetochores (one per chromatid). Therefore:
24 chromosomes x 2 kinetochores per chromosome = 48 kinetochores.
24chromosomesx2kinetochores per chromosome=48kinetochores.

125) **Explain Question –** The question asks you to interpret the chromosome diagram and correctly determine:Number of homologous chromosome pairs,Number of chromatids,Number of centromeres

Concept –cell division and cell cycle

Solution :

1. Number of homologous pairs

There are 6 chromosomes total: 3 dark ,3 light
They appear in matching pairs (same size/shape) Homologous chromosome pairs = 3

2. Number of chromatids

The diagram shows that all chromosomes are duplicated (each is X- or V-shaped → 2 chromatids each).

So,6 duplicated chromosomes × 2 chromatids = 12 chromatids Number of chromatids = 12

3. Number of centromeres

Each duplicated chromosome has one centromere.So: 6 chromosomes = 6 centromeres

Number of centromeres = 6
Final answer – 2. 3 pairs, 12 chromatids, 6 centromeres

126) NCERT XIth Pg.#168

127)

NCERT-XI, Pg. # 127, 128, Fig 10.3 & 10.4

128) NCERT-XI Pg. # 127

129) NCERT-XI, Pg # 164, 10.2

130) NCERT XII Page No. # 121, 122 [New]

131)

NCERT-XI, Pg. # 126

132) NCERT Pg. No. 127

133) NCERT Pg. # 126

134)

NCERT-XI, Pg. # 124

135) NCERT XI Page No. 164

AIM : FREE EDUCATION TO ALL APUL

136) NCERT, Pg. # 187

137) NCERT Pg. # 186

138) NCERT Pg. # 190

139) NCERT, Pg. # 187, 190

140) NCERT, Pg. # 189

141) NCERT Pg.# 190, 191

142) NCERT Pg. # 184, 186, 188, 189

143) NCERT Pg. # 187

144) Module

145) NCERT Pg. No. # 183

146)

NCERT XI, Pg # 184

147) NCERT Pg. # 189

148) NCERT Pg. 188

149) NCERT Pg. # 269

150) NCERT Pg. # 193,204

151)

NCERT Pg. # 197

152)

NCERT Pg. # 201

AIM : FREE EDUCATION TO ALL ❤ APUL

153) NCERT, Pg. # 193, 194

154) NCERT Pg#200

155) NCERT, Pg. # 202

156) NCERT Class-XII, Page No. 195, 196

157) NCERT Pg # 194

158)

NCERT Pg. # 202

159) NCERT XI Pg. # 193

160) NCERT Pg. # 194

161) NCERT Pg. # 196

162) NCERT_Pg. No. # 198

163) Module

164) NCERT XI Pg. # 200

165)

NCERT Pg. # 200

166) NCERT_Pg. No. 203

167) NCERT Pg. # 201

168)

NCERT Pg. # 202

169) NCERT-XI, Pg. # 207

AIM : FREE EDUCATION TO ALL ❤ APUL

170)

NCERT Pg. # 213

171) NCERT, Pg. # 208-212

172) NCERT XI_Pg. No. 208

173) NCERT Pg. # 214

174) NCERT, Pg. # 208-212

175) NCERT-XI, Pg. # 209

176) NCERT Pg # 205

177) NCERT Pg.# No. 206, 207

178) NCERT XI Pg. # 298

179) NCERT-XI, Pg # 298

180) NCERT Pg # 208

AIM : FREE EDUCATION TO ALL APUL